

# Reductive groups of semisimple rank 1

SGA 3, Exposé XX

English draft

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## About this draft

This is an unofficial English translation of Exposé XX, “Reductive groups of semisimple rank 1” from *Schémas en groupes* (SGA 3). The source is the recomposition edited by P. Gille and P. Polo, PDF Exp20-13oct24.pdf (35 pages), from <https://webusers.imj-prg.fr/~patrick.polo/SGA3/>. The seminar was directed by M. Demazure and A. Grothendieck (IHÉS, 1962–1964).

Statement numbers follow that PDF. A clear typographical slip is corrected in the English and marked with a % `typo:` comment; such slips are listed in the README. A slip that would change the mathematics is left as printed and marked % `typo?:`. Scholarly proofreading remains outstanding.

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# Exposé XX. Reductive groups of semisimple rank 1

by M. Demazure

## 1. Elementary systems. The groups $U_\alpha$ and $U_{-\alpha}$

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**Reminder 1.1.** Let  $S = \text{Spec}(k)$ , where  $k$  is an algebraically closed field, and let  $G$  be a reductive  $S$ -group of semisimple rank 1,  $T$  a maximal torus (necessarily split) of  $G$ . One then has

$$\mathfrak{g} = \mathfrak{t} \oplus \mathfrak{g}^\alpha \oplus \mathfrak{g}^{-\alpha},$$

where  $\alpha$  and  $-\alpha$  are the roots of  $G$  with respect to  $T$ . Moreover, there exist two monomorphisms of groups

$$p_\alpha : \mathbf{G}_{a,S} \longrightarrow G \quad \text{and} \quad p_{-\alpha} : \mathbf{G}_{a,S} \longrightarrow G$$

such that

$$tp_\alpha(x)t^{-1} = p(\alpha(t)x) \quad \text{and} \quad tp_{-\alpha}(x)t^{-1} = p_{-\alpha}(\alpha(t)^{-1}x),$$

for every  $S' \longrightarrow S$  and all  $t \in T(S')$ ,  $x \in \mathbf{G}_a(S')$ , and such that the morphism

$$\mathbf{G}_{a,S} \times_S T \times_S \mathbf{G}_{a,S} \longrightarrow G,$$

defined by  $(y, t, x) \longmapsto p_{-\alpha}(y)tp_\alpha(x)$ , is radicial and dominant (*Bible*, § 13.4, cor. 2 to th. 3).

Since the tangent map at the identity element is bijective, this morphism is also separable, hence birational; by Zariski's "Main Theorem" (EGA III<sub>1</sub>, 4.4.9), it is therefore an open immersion.

**Lemma 1.2.** Let  $S$  be a scheme,  $G$  a group  $S$ -scheme,  $T$  a torus of  $G$ ,  $Q$  a subtorus of  $T$ ,  $\alpha$  a character of  $T$  inducing on  $Q$  a character nontrivial on each fiber. Let  $p_\alpha : \mathbf{G}_{a,S} \longrightarrow G$  (resp.  $p_{-\alpha} : \mathbf{G}_{a,S} \longrightarrow G$ ) be a morphism of groups normalized by  $T$  with multiplier  $\alpha$  (resp.  $-\alpha$ ). Suppose that the morphism

$$u : \mathbf{G}_{a,S} \times_S T \times_S \mathbf{G}_{a,S} \longrightarrow G$$

defined set-theoretically by  $u(y, t, x) = p_{-\alpha}(y)tp_\alpha(x)$  is an open immersion. Finally, let  $q$  be an integer  $\geq 0$  and

$$p : \mathbf{G}_{a,S} \longrightarrow G$$

a morphism of groups such that for every  $S' \longrightarrow S$  and all  $t \in Q(S')$ ,  $x \in \mathbf{G}_a(S')$ , one has

$$\text{int}(t)^q(p(x)) = p(\alpha(t)x).$$

There then exists a unique  $\nu \in \mathbf{G}_a(S)$  such that  $p(x) = p_\alpha(\nu x^q)$ .

Indeed, let  $\Omega$  be the image of  $u$  and  $U = p^{-1}(\Omega)$ . This is an open subset of  $\mathbf{G}_{a,S}$  containing the zero section. For every section  $t$  of  $Q$ , the automorphism of  $\mathbf{G}_{a,S}$  defined by multiplication by  $\alpha(t)$  preserves  $U$  globally. One has  $U = \mathbf{G}_{a,S}$ ; indeed, it suffices to check this when  $S$  is the spectrum of an algebraically closed field  $k$ ; then  $\alpha : Q(k) \longrightarrow k^*$  is surjective, which proves at once that  $U(k) \supset k^*$ , hence  $U = \mathbf{G}_{a,k}$ . There therefore exist three morphisms

$$a : \mathbf{G}_{a,S} \longrightarrow \mathbf{G}_{a,S}, \quad b : \mathbf{G}_{a,S} \longrightarrow T, \quad c : \mathbf{G}_{a,S} \longrightarrow \mathbf{G}_{a,S},$$

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(0)**N.D.E.** Version of 13/10/2024

such that

$$p(x) = p_{-\alpha}(a(x))b(x)p_{\alpha}(c(x)).$$

The condition on  $p$  becomes

$$\begin{aligned} a(\alpha(t)x) &= \alpha(t)^{-q}a(x), \\ b(\alpha(t)x) &= b(x), \\ c(\alpha(t)x) &= \alpha(t)^q c(x). \end{aligned}$$

For the same reason as above, for every  $S' \rightarrow S$  and every  $z \in \mathbf{G}_m(S')$ , one therefore has

$$a(zx) = z^{-q}a(x), \quad b(zx) = b(x), \quad c(zx) = z^q c(x),$$

hence

$$z^q a(z) = a(1), \quad b(z) = b(1), \quad c(z) = z^q c(1).$$

Since  $\mathbf{G}_{m,S}$  is schematically dense in  $\mathbf{G}_{a,S}$ , one has at once, for every  $x \in \mathbf{G}_a(S')$ ,  $S' \rightarrow S$ :

$$\begin{aligned} x^q a(x) &= a(1) = a(0) = 0, & \text{whence } a &= 0, \\ c(x) &= x^q c(1) = \nu x^q, & \text{for some } \nu &\in \mathbf{G}_a(S), \\ b(x) &= b(1) = b(0) = e, & \text{whence } b &= e, \end{aligned}$$

which completes the proof.

**Definition 1.3.** Let  $S$  be a scheme. An *elementary  $S$ -system* is a triple  $(G, T, \alpha)$  where

- (i)  $G$  is a reductive  $S$ -group of semisimple rank 1 (Exp. XIX 2.7),
- (ii)  $T$  is a maximal torus of  $G$ ,
- (iii)  $\alpha$  is a root of  $G$  with respect to  $T$  (Exp. XIX 3.2).

One therefore has a direct sum decomposition (Exp. XIX 3.5)<sup>(1)</sup>

$$\mathfrak{g} = \mathfrak{t} \oplus \mathfrak{g}^{\alpha} \oplus \mathfrak{g}^{-\alpha},$$

$\mathfrak{g}^{\alpha}$  and  $\mathfrak{g}^{-\alpha}$  being locally free of rank one.

**1.4.** — If  $(G, T, \alpha)$  is an elementary  $S$ -system, then  $(G_{S'}, T_{S'}, \alpha_{S'})$  is an elementary  $S'$ -system for every  $S' \rightarrow S$ . If  $(G, T, \alpha)$  is an elementary  $S$ -system, then  $(G, T, -\alpha)$  is also one.

If  $S$  is a scheme,  $G$  a reductive  $S$ -group,  $T$  a maximal torus of  $G$ ,  $\alpha$  a root of  $G$  with respect to  $T$ , then (Exp. XIX 3.9)  $(Z_{\alpha}, T, \alpha)$  is an elementary  $S$ -system.

Let  $(G, T, \alpha)$  be an elementary  $S$ -system. The invertible module  $\mathfrak{g}^{\alpha}$  is canonically equipped with a  $T$ -module structure. One therefore also has a  $T$ -module structure on the vector bundle  $W(\mathfrak{g}^{\alpha})$ . On the other hand, the inner automorphisms of  $T$  define on  $G$  a structure of group with operator group  $T$ .

**Theorem 1.5.** Let  $(G, T, \alpha)$  be an elementary  $S$ -system.

- (i) There exists a unique morphism of groups with operator group  $T$

$$\exp : W(\mathfrak{g}^{\alpha}) \rightarrow G$$

which induces on Lie algebras the canonical morphism  $\mathfrak{g}^{\alpha} \rightarrow \mathfrak{g}$ .<sup>(2)</sup>

In other words,  $\exp$  is the unique morphism satisfying the following conditions: for every  $S' \rightarrow S$  and all  $t \in T(S')$ ,  $X, X' \in W(\mathfrak{g}^{\alpha})(S')$ , one has

$$\begin{aligned} \exp(X + X') &= \exp(X) \exp(X'), \\ \text{int}(t)(\exp(X)) &= \exp(\alpha(t)X), \\ \underline{\text{Lie}}(\exp)(X) &= X. \end{aligned}$$

<sup>(1)</sup>**N.D.E.** Of  $\mathcal{O}_S$ -modules.

<sup>(2)</sup>**N.D.E.** We shall see later (Cor. 5.9) that  $\exp$  is an isomorphism from  $W(\mathfrak{g}^{\alpha})$  onto a closed subgroup of  $G$ .

(ii) If one defines similarly (in the elementary  $S$ -system  $(G, T, -\alpha)$ )

$$\exp : W(\mathfrak{g}^{-\alpha}) \longrightarrow G,$$

then the morphism

$$W(\mathfrak{g}^{-\alpha}) \times_S T \times_S W(\mathfrak{g}^{\alpha}) \longrightarrow G$$

defined set-theoretically by  $(Y, t, X) \mapsto \exp(Y) \cdot t \cdot \exp(X)$  is an open immersion.

Suppose we have proved the existence of the required morphisms  $\exp$ , and let us prove the other assertions of the theorem. Let us first prove (ii). Since both sides are of finite presentation and flat over  $S$ , it suffices to do so when  $S$  is the spectrum of an algebraically closed field (SGA 1, I 5.7 and VIII 5.5). Let then  $S = \operatorname{Spec} k$ . Let  $X \in \Gamma(S, \mathfrak{g}^{\alpha})^{\times}$ ,  $Y \in \Gamma(S, \mathfrak{g}^{-\alpha})^{\times}$ . It suffices to prove that the morphism

$$\mathbf{G}_{a,k} \times_k T \times_k \mathbf{G}_{a,k} \longrightarrow G, \quad (y, t, x) \mapsto \exp(yY)t \exp(xX)$$

is an open immersion. Now, by 1.1 and 1.2, there exist  $a, b \in k$  such that

$$\exp(yY) = p_{-\alpha}(ay) \quad \text{and} \quad \exp(xX) = p_{\alpha}(bx).$$

Since  $\exp : W(\mathfrak{g}^{-\alpha}) \longrightarrow G$  induces a monomorphism on Lie algebras, one has  $a \neq 0$ ; similarly  $b \neq 0$ , and one is reduced to 1.1.

The uniqueness of the morphism  $\exp$  can be proved locally on  $S$ ; one then reduces to the case where  $\mathfrak{g}^{\alpha}$  and  $\mathfrak{g}^{-\alpha}$  are free, and one need only apply 1.2 (with  $Q = T$  and  $q = 1$ ).

It therefore remains to prove the existence of the required morphism  $\exp$ . Note first that, by the theory of faithfully flat descent and the preceding uniqueness assertion, it suffices to prove this existence locally on  $S$  for the (fpqc) topology. By the usual arguments using finite presentation, one reduces to the case where  $S$  is noetherian, then to the case where it is local noetherian. By the preceding remark, one may therefore content oneself with proving the existence of the desired morphism  $\exp$  when  $S = \operatorname{Spec}(A)$ ,  $A$  local noetherian complete with algebraically closed residue field  $k$ . Let then  $p_0 : \mathbf{G}_{a,k} \longrightarrow G_k$  be a monomorphism of  $k$ -groups normalized by  $T_k$  with multiplier  $\alpha_0 = \alpha \otimes_A k$  (one exists by 1.1). One knows (1.1 and 1.2) that the corresponding morphism  $T_k \cdot_{\alpha_0} \mathbf{G}_{a,k} \longrightarrow G_k$  is an immersion, hence in particular a monomorphism. Let us provisionally admit the following two lemmas:

**Lemma 1.6.** *Let  $S$  be a scheme,  $G$  an  $S$ -group of finite presentation,  $T$  an  $S$ -torus,  $\alpha$  a character nontrivial on each fiber of  $T$ ,  $s_0$  a point of  $S$ . Let*

$$f : T \cdot_{\alpha} \mathbf{G}_{a,S} \longrightarrow G$$

*be a morphism of  $S$ -groups such that  $f_{s_0}$  is a monomorphism and the restriction of  $f$  to  $T$  is a monomorphism. There exists an open neighborhood  $U$  of  $s_0$  such that  $f|_U$  is a monomorphism.*

**Lemma 1.7.** *Let  $A$  be a complete local noetherian ring with algebraically closed residue field  $k$ ,  $(G, T, \alpha)$  an elementary  $A$ -system,  $p_0 : \mathbf{G}_{a,k} \longrightarrow G_k$  a morphism of  $k$ -groups normalized by  $T_k$  with multiplier  $\alpha \otimes_A k$ . There exists a morphism of groups  $p : \mathbf{G}_{a,A} \longrightarrow G$  normalized by  $T$  with multiplier  $\alpha$ .*

Let  $p$  be the morphism whose existence is asserted by 1.7. Let  $f : T \cdot_{\alpha} \mathbf{G}_{a,S} \longrightarrow G$  be the corresponding morphism. It satisfies the hypotheses of 1.6, hence is a monomorphism; in particular  $p$  is a monomorphism. One then concludes by Exp. XIX 4.9.

*Proof of 1.6.* Let  $\varepsilon : S \longrightarrow T \cdot_{\alpha} \mathbf{G}_{a,S}$  denote the unit section. Since  $f$  is unramified at  $\varepsilon(s_0)$ , it is so at  $\varepsilon(s)$  for all  $s$  in an open neighborhood  $U$  of  $s_0$ ;  $f|_U$  is therefore unramified (Exp. X 3.5),<sup>(3)</sup> hence its kernel  $\operatorname{Ker}(f)|_U$  is unramified over  $U$ . To prove that  $f|_U$  is a monomorphism, it therefore suffices<sup>(4)</sup> to prove that  $\operatorname{Ker}(f)|_U$  is radicial over  $U$ , which is a set-theoretic question. One is therefore reduced to proving:

<sup>(3)</sup>**N.D.E.** See also VI<sub>B</sub>, 2.5.

<sup>(4)</sup>**N.D.E.** By EGA IV<sub>4</sub>, 17.9.1.

**Lemma 1.8.** *Let  $k$  be an algebraically closed field; let  $N$  be an invariant subgroup of  $T \cdot_\alpha \mathbf{G}_{a,k}$  ( $\alpha$  a nontrivial character of the torus  $T$ ), étale over  $k$  and such that  $N \cap T = \{e\}$ . Then  $N = \{e\}$ .*

One has  $\text{int}(t')(t, x) = (t, \alpha(t')x)$ . If  $(t, x)$  is a point of  $N$ , with  $x \neq 0$ , then  $(t, zx)$  is also a point of  $N$  for  $z \in k^*$ , and  $(t, x)$  is not isolated, hence  $N$  is not quasi-finite. One therefore has set-theoretically  $N \subset T$ , and the proof is finished.

*Proof of 1.7.* Let  $\mathfrak{m}$  be the radical of  $A$  and  $S_n = \text{Spec}(A/\mathfrak{m}^{n+1})$ ,  $n \geq 0$ . Let us first show, by induction on  $n$ , that  $p_0$  can be extended for each  $n$  to a morphism of  $S_n$ -groups

$$p_n : \mathbf{G}_{a,S_n} \longrightarrow G_{S_n}$$

normalized by  $T_{S_n}$  with multiplier  $\alpha_{S_n}$ , the  $p_n$  satisfying in addition  $p_{n+1} \times_{S_{n+1}} S_n = p_n$ .

Let  $H = T \cdot_\alpha \mathbf{G}_{a,S}$ . The morphism  $H_{S_n} \longrightarrow G_{S_n}$  defined by  $p_n$  is denoted  $f_n$ . Let us admit the following lemma:

**Lemma 1.9.** *If  $(G, T, \alpha)$  is an elementary  $k$ -system,  $k$  algebraically closed, and  $p : \mathbf{G}_{a,k} \longrightarrow G$  is a monomorphism normalized by  $T$  with multiplier  $\alpha$ , one has*

$$H^2(T \cdot_\alpha \mathbf{G}_{a,k}, \mathfrak{g}) = 0.$$

(One makes  $T \cdot_\alpha \mathbf{G}_{a,k}$  act on  $\mathfrak{g}$  through the morphism  $T \cdot_\alpha \mathbf{G}_{a,k} \longrightarrow G$  defined by  $p$ , and the adjoint representation of  $G$ .)

Then, by Exp. III 2.8,  $f_n$  will extend to a morphism of  $S_{n+1}$ -groups

$$f'_{n+1} : H_{S_{n+1}} \longrightarrow G_{S_{n+1}}.$$

Now  $f'_{n+1}$  and the canonical immersion of  $T_{S_{n+1}}$  into  $G_{S_{n+1}}$  have the same restriction to  $T_{S_n}$ . By Exp. III 2.5, there exists an element  $g \in G(S_{n+1})$  such that  $g \times_{S_{n+1}} S_n = e$  and such that  $f_{n+1} = \text{int}(g) \circ f'_{n+1}$  restricts to  $T_{n+1}$  by the canonical immersion of  $T_{n+1}$ . Let  $p_{n+1}$  be the restriction of  $f_{n+1}$  to  $\mathbf{G}_{a,S_{n+1}}$ . It is a morphism normalized by  $T_{S_{n+1}}$  with multiplier  $\alpha_{S_{n+1}}$ , extending  $p_n$ .

One has therefore constructed a coherent system  $(f_n)$ , and one must now algebraize it. Now one has:

**Lemma 1.10.** *Let  $A$  be a complete local noetherian ring,  $\mathfrak{m}$  its maximal ideal,  $S = \text{Spec}(A)$ ,  $S_n = \text{Spec}(A/\mathfrak{m}^{n+1})$ ,  $T$  an  $S$ -torus,  $\alpha$  a nonzero character of  $T$ ,  $X$  an affine  $S$ -scheme on which  $T$  acts. Let  $T$  act on  $\mathbf{G}_{a,S}$  through  $\alpha$ . Let  $q$  be an integer  $\geq 0$ , and let  $(f_n)_{n \geq 0}$  be a coherent system of morphisms*

$$f_n : \mathbf{G}_{a,S_n}^q \longrightarrow X_{S_n}$$

*of objects with operators  $T_{S_n}$ . There exists a unique morphism of objects with operators  $T$*

$$f : \mathbf{G}_{a,S}^q \longrightarrow X$$

*inducing the  $f_n$  (compare with Exp. IX 7.1).*

**Corollary 1.11.** *If  $X$  is a group with operator group  $T$  and the  $f_n$  are morphisms of groups,  $f$  is also one.*

It suffices to apply the uniqueness assertion of the lemma to the two morphisms  $\mathbf{G}_{a,S}^{2q} \longrightarrow X$  deduced from  $f$  in the usual way.

*Proof of 1.10.* Suppose  $T$  split, which is moreover the case in the application of 1.10 to the proof of 1.5. One knows (Exp. I 4.7.3.1) that  $X \longmapsto \mathcal{A}(X)$  gives an equivalence between the category of affine  $S$ -schemes equipped with an action of  $T$  and the opposite of the category of  $S$ -algebras graded of type  $M = \text{Hom}_{S\text{-gr.}}(T, \mathbf{G}_{m,S})$ .

One therefore has gradings

$$B = \mathcal{A}(X) = \coprod_{m \in M} B_m \quad \text{and} \quad C = \mathcal{A}(\mathbf{G}_{a,S}^q) = \coprod_{m \in M} C_m.$$

One sees at once that each  $C_m$  is *free of finite type* over  $A$ . (Indeed,  $C_m = 0$  if  $m$  is not a multiple of  $\alpha$ , and if  $m = d\alpha$ ,  $C_m$  is isomorphic to the  $A$ -module of homogeneous polynomials of degree  $d$  in  $q$  variables.) Put

$$\begin{aligned}\widehat{B}_m &= \varprojlim_n B_m \otimes_A (A/\mathfrak{m}^{n+1}), \\ \widehat{C}_m &= \varprojlim_n C_m \otimes_A (A/\mathfrak{m}^{n+1}), \\ \widehat{B} &= \prod_{m \in M} \widehat{B}_m, \quad \widehat{C} = \prod_{m \in M} \widehat{C}_m.\end{aligned}$$

One then has canonical morphisms of algebras graded of type  $M$

$$g_B : B \longrightarrow \widehat{B} \quad \text{and} \quad g_C : C \longrightarrow \widehat{C}.$$

It follows from the remark above that  $g_C$  is an isomorphism. Giving a coherent system  $(f_n)$  as in the statement is equivalent to giving a morphism of graded  $A$ -algebras

$$\widehat{F} : \widehat{B} \longrightarrow \widehat{C}.$$

Finding a morphism  $f$  as in the statement is equivalent to finding a morphism of graded  $A$ -algebras  $F : B \longrightarrow C$  making the diagram commute:

$$\begin{array}{ccc} B & \xrightarrow{F} & C \\ g_B \downarrow & & \downarrow g_C \\ \widehat{B} & \xrightarrow{\widehat{F}} & \widehat{C}. \end{array}$$

Since  $g_C$  is an isomorphism, the existence and uniqueness of  $F$  are immediate. This proves 1.10.

To finish the proof of 1.5, it therefore remains only to prove 1.9.

**1.12.** — *Proof of 1.9.* One has  $\mathfrak{g} = \mathfrak{t} \oplus \mathfrak{g}^\alpha \oplus \mathfrak{g}^{-\alpha}$ . As explained in 1.9, consider  $\mathfrak{g}$  as a  $(T \cdot_\alpha \mathbf{G}_{a,k})$ -module. Clearly  $\mathfrak{t} \oplus \mathfrak{g}^\alpha$  is a submodule of  $\mathfrak{g}$ , the quotient being isomorphic to  $\mathfrak{g}^{-\alpha}$  as a  $k$ -vector space and even as a  $T$ -module. Clearly  $\mathbf{G}_{a,k}$  acts trivially on this one-dimensional quotient (since every group morphism from  $\mathbf{G}_{a,k}$  to  $\mathbf{G}_{m,k}$  is trivial). Similarly,  $\mathfrak{g}^\alpha$  is a submodule of  $\mathfrak{t} \oplus \mathfrak{g}^\alpha$ , the quotient being isomorphic to  $\mathfrak{t}$  as a  $T$ -module,  $\mathbf{G}_{a,k}$  acting trivially on it. To summarize:

**Lemma 1.13.** *Under the conditions of 1.9,  $\mathfrak{g}$  has a composition series as a  $(T \cdot_\alpha \mathbf{G}_{a,k})$ -module whose successive quotients are*

$$\mathfrak{g}^{-\alpha}, \quad \mathfrak{t}, \quad \mathfrak{g}^\alpha,$$

*considered as  $(T \cdot_\alpha \mathbf{G}_{a,k})$ -modules by the projection  $T \cdot_\alpha \mathbf{G}_{a,k} \longrightarrow T$ .*

One is therefore reduced to calculating the cohomology of  $T \cdot_\alpha \mathbf{G}_{a,k}$  acting through the projection  $T \cdot_\alpha \mathbf{G}_{a,k} \longrightarrow T$  and the character  $\beta$  of  $T$  (here  $\beta = 0, \alpha$  or  $-\alpha$ ) on  $W(k)$ .<sup>(5)</sup> Denote by  $k[x_1, \dots, x_n]$  the polynomial algebra over  $k$  in  $n$  variables, and by  $k_q[x_1, \dots, x_n]$  the subspace of homogeneous polynomials of degree  $q$ .

**Lemma 1.14.** *With the preceding notations, one has  $H^n(T \cdot_\alpha \mathbf{G}_{a,k}, k) = H^n(C_{\alpha, \beta}^*)$ , where the complex  $C_{\alpha, \beta}^*$  is defined by*

$$C_{\alpha, \beta}^n = \begin{cases} k_q[x_1, \dots, x_n] & \text{if } \beta = q\alpha, \text{ with } q \in \mathbf{N}^*; \\ 0 & \text{otherwise,} \end{cases}$$

<sup>(5)</sup>**N.D.E.** The following sentence has been added.

and

$$\begin{aligned} \delta f(x_1, x_2, \dots, x_{n+1}) &= f(x_2, \dots, x_{n+1}) \\ &+ \sum_{i=1}^n (-1)^i f(x_1, \dots, x_i x_{i+1}, \dots, x_{n+1}) \\ &+ (-1)^{n+1} f(x_1, \dots, x_n). \end{aligned}$$

Indeed, the functor  $M \mapsto H^0(T, M)$  is exact on the category of  $T$ -modules (and the  $H^q(T, -)$  vanish), by Exp. I 5.3.2. It follows, as in the usual case of group cohomology, that  $H^n(T \cdot_\alpha \mathbf{G}_{a,k}, k)$  can be calculated as the  $n$ th cohomology group of the complex of cochains from  $\mathbf{G}_{a,k}$  to  $k$ , invariant under  $T$ , i.e. satisfying

$$f(\alpha(t)x_1, \dots, \alpha(t)x_n) = \beta(t)f(x_1, \dots, x_n).$$

This indeed gives the announced complex.

To prove 1.9, it therefore suffices to prove that  $H^2(C_{\alpha,\beta}^*) = 0$  for  $\beta = 0, \alpha, -\alpha$ , which is immediate.

**Remark 1.15.** One can explicitly calculate the groups  $H^n(C_{\alpha,\beta}^*)$  for  $\beta = q\alpha$  (see M. Lazard, *Lois de groupes et analyseurs*, Annales E.N.S., 1955). In particular, one finds  $H^n(C_{\alpha,q\alpha}^*) = 0$  for  $n > q$ .

**Notation 1.16.** The image of the canonical immersion

$$W(\mathfrak{g}^{-\alpha}) \times_S T \times_S W(\mathfrak{g}^\alpha) \longrightarrow G$$

will be denoted  $\Omega$ . It is an open subset of  $G$  containing the unit section. The image of

$$W(\mathfrak{g}^{-\alpha}), \quad \text{resp. } W(\mathfrak{g}^\alpha), \quad \text{resp. } W(\mathfrak{g}^{-\alpha}) \times_S T, \quad \text{resp. } T \times_S W(\mathfrak{g}^\alpha)$$

will be denoted<sup>(6)</sup>

$$U_{-\alpha}, \quad \text{resp. } U_\alpha, \quad \text{resp. } U_{-\alpha} \cdot T, \quad \text{resp. } T \cdot U_\alpha.$$

Then  $U_\alpha$  (resp.  $U_{-\alpha}$ ) is a subgroup of  $G$  canonically equipped with a vector bundle structure, and one has

$$\text{int}(t)(x) = x^{\alpha(t)} \quad (\text{resp. } x^{-\alpha(t)}),$$

for all  $S' \longrightarrow S$ ,  $t \in T(S')$ ,  $x \in U_\alpha(S')$  (resp.  $x \in U_{-\alpha}(S')$ ).

One has canonical isomorphisms

$$T \cdot U_\alpha \simeq T \cdot_\alpha U_\alpha \quad \text{and} \quad T \cdot U_{-\alpha} \simeq T \cdot_{-\alpha} U_{-\alpha}.$$

The open subset  $\Omega$  is stable under  $\text{int}(T)$ : one has

$$\text{int}(t')(y \cdot t \cdot x) = y^{-\alpha(t')} \cdot t \cdot x^{\alpha(t')}.$$

**Corollary 1.17.** *One has  $\mathcal{L}ie(U_\alpha/S) = \mathfrak{g}^\alpha$  and  $\mathcal{L}ie(U_{-\alpha}/S) = \mathfrak{g}^{-\alpha}$ . The isomorphisms*

$$W(\mathfrak{g}^\alpha) \xrightarrow[\sim]{\exp} U_\alpha \quad \text{and} \quad W(\mathfrak{g}^{-\alpha}) \xrightarrow[\sim]{\exp} U_{-\alpha}$$

*are those of Exp. XIX 4.2.*

**Corollary 1.18.** *The open subset  $\Omega$  is relatively schematically dense in  $G$  (cf. XVIII, § 1).*

Clear by Exp. XVIII, 1.3.<sup>(7)</sup>

**Corollary 1.19.** *The center of  $G$  is  $\text{Centr}(G) = \text{Ker}(\alpha)$ . It is therefore a closed subgroup of  $G$ , of multiplicative type and of finite type.*

The second assertion follows from the first by Exp. IX 2.7. Let us therefore prove the latter. The inner automorphism defined by a section of  $\text{Ker}(\alpha)$  acts trivially on  $\Omega$  (last

<sup>(6)</sup>**N.D.E.**  $P_{-\alpha}$  and  $P_\alpha$  have been replaced by  $U_{-\alpha}$  and  $U_\alpha$ .

<sup>(7)</sup>**N.D.E.** See also EGA IV<sub>3</sub>, 11.10.10.

formula of 1.16), hence on  $G$  by 1.18. Conversely, if  $g \in G(S)$  centralizes  $G$ , it centralizes  $T$  and  $U_\alpha$ , hence is a section of  $T$  (Exp. XIX 2.8) annihilating  $\alpha$ ; since this also holds after every base change, one indeed has  $\text{Centr}(G) = \text{Ker}(\alpha)$ .

**Corollary 1.20.** *For there to exist a monomorphism  $p_\alpha : \mathbf{G}_{a,S} \rightarrow G$  normalized by  $T$  with multiplier  $\alpha$ , it is necessary and sufficient that the  $\mathcal{O}_S$ -module  $\mathfrak{g}^\alpha$  be free. More precisely, one has a bijection given by*

$$X_\alpha \mapsto (x \mapsto \exp(xX_\alpha)) \quad \text{and} \quad p_\alpha \mapsto \text{Lie}(p_\alpha)$$

between  $\Gamma(S, \mathfrak{g}^\alpha)^\times$  and the set of monomorphisms  $p_\alpha$  as above (which is also the set of isomorphisms of vector group schemes  $\mathbf{G}_{a,S} \xrightarrow{\sim} U_\alpha$ ).<sup>(8)</sup>

**Corollary 1.21.** *The subgroups  $U_\alpha$  and  $U_{-\alpha}$  of  $G$  do not commute on any fiber.*

Indeed, if  $(U_\alpha)_s$  and  $(U_{-\alpha})_s$  commute,  $\Omega_s$  is a subgroup of  $G_s$ , hence  $\Omega_s = G_s$ <sup>(9)</sup> and  $G_s$  is solvable, which contradicts the hypothesis that  $G_s$  is reductive of semisimple rank 1.

## 2. Structure of elementary systems

**Theorem 2.1.** *Let  $S$  be a scheme,  $(G, T, \alpha)$  an elementary  $S$ -system. There exist a morphism of  $\mathcal{O}_S$ -modules*

$$\mathfrak{g}^\alpha \otimes_{\mathcal{O}_S} \mathfrak{g}^{-\alpha} \rightarrow \mathcal{O}_S, \quad (X, Y) \mapsto \langle X, Y \rangle,$$

and a morphism of  $S$ -groups

$$\alpha^* : \mathbf{G}_{m,S} \rightarrow T$$

such that for every  $S' \rightarrow S$  and all  $X \in \Gamma(S', \mathfrak{g}^\alpha \otimes \mathcal{O}_{S'})$ ,  $Y \in \Gamma(S', \mathfrak{g}^{-\alpha} \otimes \mathcal{O}_{S'})$  one has the equivalence

$$\exp(X) \cdot \exp(Y) \in \Omega(S') \iff 1 + \langle X, Y \rangle \in \mathbf{G}_m(S'),$$

and under these conditions one has the formula:

$$(F) \quad \boxed{\exp(X) \cdot \exp(Y) = \exp\left(\frac{Y}{1 + \langle X, Y \rangle}\right) \alpha^*(1 + \langle X, Y \rangle) \exp\left(\frac{X}{1 + \langle X, Y \rangle}\right).}$$

Moreover, the morphisms  $(X, Y) \mapsto \langle X, Y \rangle$  and  $\alpha^*$  are uniquely determined, the first is an isomorphism, hence puts the modules  $\mathfrak{g}^\alpha$  and  $\mathfrak{g}^{-\alpha}$  in duality, and one has  $\alpha \circ \alpha^* = 2$  (squaring in  $\mathbf{G}_{m,S}$ ).

In view of the uniqueness assertions of the theorem, it suffices to give the proof locally on  $S$ . One may therefore suppose that  $\mathfrak{g}^\alpha$  and  $\mathfrak{g}^{-\alpha}$  are free over  $S$ . Take then  $X \in \Gamma(S, \mathfrak{g}^\alpha)^\times$ ,  $Y \in \Gamma(S, \mathfrak{g}^{-\alpha})^\times$ , and put  $p_\alpha(x) = \exp(xX)$ ,  $p_{-\alpha}(y) = \exp(yY)$  for  $x, y \in \mathbf{G}_a(S')$ ,  $S' \rightarrow S$ . By 1.5 and 1.21, it suffices to prove:

**Proposition 2.2.** *Let  $S$  be a scheme,  $G$  an  $S$ -group,  $T$  a torus of  $G$ ,  $\alpha$  a character of  $T$  nontrivial on each fiber,  $p_\alpha : \mathbf{G}_{a,S} \rightarrow G$  (resp.  $p_{-\alpha} : \mathbf{G}_{a,S} \rightarrow G$ ) a monomorphism of groups normalized by  $T$  with multiplier  $\alpha$  (resp.  $-\alpha$ ). Suppose that:*

- (i) *The morphism  $\mathbf{G}_{a,S} \times_S T \times_S \mathbf{G}_{a,S} \rightarrow G$  defined by  $(y, t, x) \mapsto p_{-\alpha}(y)tp_\alpha(x)$  is an open immersion. Denote its image by  $\Omega$ .*
- (ii) *For every  $s \in S$ ,  $(p_\alpha)_s(\mathbf{G}_{a,\kappa(s)})$  and  $(p_{-\alpha})_s(\mathbf{G}_{a,\kappa(s)})$  do not commute.*

*Then there exist uniquely determined  $a \in \mathbf{G}_a(S)$  and  $\alpha^* \in \text{Hom}_{S\text{-gr.}}(\mathbf{G}_{m,S}, T)$  with the following properties: for every  $S' \rightarrow S$  and all  $x, y \in \mathbf{G}_a(S')$ , one has*

$$p_\alpha(x)p_{-\alpha}(y) \in \Omega(S') \iff 1 + axy \in \mathbf{G}_m(S'),$$

<sup>(8)</sup>**N.D.E.** Indeed, on the one hand,  $\text{Lie}(\mathbf{G}_{a,S}) = \mathcal{O}_S$ , and  $\text{Lie}(p_\alpha)$  is an element of  $\text{Hom}_{\mathcal{O}_S}(\mathcal{O}_S, \mathfrak{g}^\alpha) = \Gamma(S, \mathfrak{g}^\alpha)$ .

<sup>(9)</sup>**N.D.E.** By 1.18.

and, under this condition, one has the formula

$$p_\alpha(x)p_{-\alpha}(y) = p_{-\alpha}\left(\frac{y}{1+axy}\right)\alpha^*(1+axy)p_\alpha\left(\frac{x}{1+axy}\right).$$

Moreover,  $a$  is invertible (i.e.  $a \in \mathbf{G}_m(S)$ ) and  $\alpha \circ \alpha^* = 2$ .

*Proof:*

**A)** Consider the morphism

$$\mathbf{G}_{a,S}^2 \longrightarrow G$$

defined by  $(x, y) \mapsto p_\alpha(x)p_{-\alpha}(y)$ . Let  $U$  be the inverse image of  $\Omega$  by this morphism. It is an open subset of  $\mathbf{G}_{a,S}^2$  containing  $0 \times_S \mathbf{G}_{a,S}$  and  $\mathbf{G}_{a,S} \times_S 0$ . There therefore exist uniquely determined morphisms of  $S$ -schemes

$$A : U \longrightarrow \mathbf{G}_{a,S}, \quad C : U \longrightarrow \mathbf{G}_{a,S}, \quad B : U \longrightarrow T$$

satisfying the set-theoretic relation:

$$p_\alpha(u)p_{-\alpha}(v) = p_{-\alpha}(A(u, v))B(u, v)p_\alpha(C(u, v)).$$

One immediately has the relations

$$\begin{aligned} A(0, v) &= v, & A(u, 0) &= 0, & C(u, 0) &= u, & C(0, v) &= 0, \\ B(u, 0) &= B(0, v) = e. \end{aligned}$$

Let  $S'$  be a separated  $S$ -scheme and  $t \in T(S')$  a point of  $T$ . Since  $\Omega_{S'}$  is stable under  $\text{int}(t)$ , by the last formula of 1.16,  $U_{S'}$  is stable under the automorphism  $(x, y) \mapsto (\alpha(t)x, \alpha(t)^{-1}y)$  of  $\mathbf{G}_{a,S'}^2$ , and one has the relations:

$$\begin{aligned} A(\alpha(t)u, \alpha(t)^{-1}v) &= \alpha(t)^{-1}A(u, v), \\ C(\alpha(t)u, \alpha(t)^{-1}v) &= \alpha(t)C(u, v), \\ B(\alpha(t)u, \alpha(t)^{-1}v) &= B(u, v). \end{aligned}$$

Since  $\alpha$  is faithfully flat, one deduces that for every  $S' \rightarrow S$  and every  $z \in \mathbf{G}_m(S')$ ,  $U_{S'}$  is stable under the transformation  $(x, y) \mapsto (zx, z^{-1}y)$  and one has

$$\begin{aligned} A(zu, z^{-1}v) &= z^{-1}A(u, v), & C(zu, z^{-1}v) &= zC(u, v), \\ B(zu, z^{-1}v) &= B(u, v). \end{aligned}$$

Suppose first that  $v$  is invertible; putting  $z = v$ , one deduces that if  $(u, v)$  is a section of  $U$ , then  $(uv, 1)$  is also one, and one has

$$A(uv, 1) = v^{-1}A(u, v), \quad B(uv, 1) = B(u, v).$$

Let then  $V$  be the open subset of  $\mathbf{G}_{a,S}^2$  defined by<sup>(10)</sup>

$$(u, v) \in V(S') \iff (u, v), (uv, 1) \text{ and } (1, uv) \text{ belong to } U(S').$$

Since  $U$  is an open subset of  $\mathbf{G}_{a,S}^2$  containing  $0 \times_S \mathbf{G}_{a,S}$  and  $\mathbf{G}_{a,S} \times_S 0$ ,  $V$  is a neighborhood of the zero section of  $\mathbf{G}_{a,S}^2$ , and we have just seen that the morphisms

$$\begin{aligned} (u, v) &\mapsto A(u, v) \quad \text{and} \quad (u, v) \mapsto vA(uv, 1), \\ \text{resp. } (u, v) &\mapsto B(u, v) \quad \text{and} \quad (u, v) \mapsto B(uv, 1) \end{aligned}$$

coincide in  $V \cap (\mathbf{G}_{a,S} \times_S \mathbf{G}_{m,S})$ . Since  $\mathbf{G}_{a,S} \times_S \mathbf{G}_{m,S}$  is schematically dense in  $\mathbf{G}_{a,S}^2$ , these morphisms therefore coincide in  $V$ .

One knows that  $A(0, 1) = 1$ ; it follows that there exists an open subset  $W_1$  of  $\mathbf{G}_{a,S}$  containing the zero section, such that for every section  $x$  of  $W_1$ ,  $A(x, 1)$  is invertible; putting  $A(x, 1)^{-1} = F(x)$ , one obtains that if  $(u, v) \in V(S')$ <sup>(11)</sup> and  $uv \in W_1(S')$ ,  $S' \rightarrow S$ , then

<sup>(10)</sup>**N.D.E.** The condition “ $(1, uv) \in U(S')$ ” has been added.

<sup>(11)</sup>**N.D.E.** Here and in what follows,  $U(S')$  has been replaced by  $V(S')$ .

$A(u, v) = vA(uv, 1) = vF(uv)^{-1}$ . Reasoning similarly for  $C$ , one obtains that there exist an open subset  $W_2$  of  $\mathbf{G}_{a,S}$  containing the zero section and an element  $E^{(12)}$  of  $\mathcal{O}(W_2)^\times$  such that  $C(u, v) = uC(1, uv) = uE(uv)^{-1}$  if  $(u, v) \in V(S')$  and  $uv \in W_2(S')$ . Consequently, putting  $W = W_1 \cap W_2$ , one obtains:

There exist an open subset  $W$  of  $\mathbf{G}_{a,S}$  containing the zero section, and  $S$ -morphisms

$$\begin{aligned} F : W &\longrightarrow \mathbf{G}_{m,S}, & F(0) &= 1, \\ H : W &\longrightarrow T, & H(0) &= e, \\ E : W &\longrightarrow \mathbf{G}_{m,S}, & E(0) &= 1, \end{aligned}$$

such that if  $(u, v) \in V(S')$  and  $uv \in W(S')$ ,  $S' \longrightarrow S$ , one has

$$(+) \quad p_\alpha(u)p_{-\alpha}(v) = p_{-\alpha}(vF(uv)^{-1})H(uv)p_\alpha(uE(uv)^{-1}).$$

**B)** Let us now use associativity in  $G$  to write

$$p_\alpha(u)p_{-\alpha}(v)p_{-\alpha}(w) = p_\alpha(u)p_{-\alpha}(v+w).$$

There exists an open subset  $L$  of  $\mathbf{G}_{a,S}^3$ , containing the unit section, such that  $(u, v, w) \in L(S')$  is equivalent to

$$\begin{aligned} (u, v) \in V(S'), \quad (uE(uv)^{-1}, w) \in V(S'), \quad (u, v+w) \in V(S'), \\ uv \in W(S'), \quad uwE(uv)^{-1} \in W(S'), \quad u(v+w) \in W(S'). \end{aligned}$$

Using formula (+), one writes at once, for  $(u, v, w) \in L(S')$ , the relations:

$$\begin{aligned} (1) \quad E(uv + uw) &= E(uwE(uv)^{-1})E(uv), \\ (2) \quad H(uv + uw) &= H(uwE(uv)^{-1})H(uv), \\ (3) \quad (v+w)F(uv + uw)^{-1} &= \alpha(H(uv)^{-1})wF(uwE(uv)^{-1})^{-1} \\ &\quad + vF(uv)^{-1}. \end{aligned}$$

It is immediate from the definition of  $L$  that  $(1, 0, 0) \in L(S)$ . Consider therefore

$$L \cap (1 \times_S \mathbf{G}_{a,S} \times_S \mathbf{G}_{a,S}) = 1 \times_S M;$$

$M$  is an open subset of  $\mathbf{G}_{a,S}^2$  containing the section  $(0, 0)$ , and, for  $(v, w) \in M(S')$ , one has  $v, wE(v)^{-1}, v+w \in W(S')$  and

$$\begin{aligned} (1') \quad E(v+w) &= E(wE(v)^{-1})E(v), \\ (2') \quad H(v+w) &= H(wE(v)^{-1})H(v), \\ (3') \quad (v+w)F(v+w)^{-1} &= \alpha(H(v))^{-1}wF(wE(v)^{-1})^{-1} + vF(v)^{-1}. \end{aligned}$$

Finally, consider the morphism from  $M$  to  $\mathbf{G}_{a,S}^2$  defined set-theoretically by  $(v, w) \mapsto (v, wE(v)^{-1})$ .<sup>(13)</sup> It preserves the section  $(0, 0)$  and induces an isomorphism from  $M$  onto an open subset  $N$  of  $\mathbf{G}_{a,S}^2$  containing the zero section (the inverse isomorphism being given by  $(x, y) \mapsto (x, yE(x))$ ).<sup>(14)</sup> One has therefore proved the following assertion:

There exists an open subset  $N$  of  $\mathbf{G}_{a,S}^2$  containing the zero section such that if  $(x, y) \in N(S')$ , then  $x, y$  and  $x + yE(x)$ <sup>(15)</sup> belong to  $W(S')$  and:

$$\begin{aligned} (1'') \quad E(x + yE(x)) &= E(x)E(y), \\ (2'') \quad H(x + yE(x)) &= H(x)H(y), \\ (3'') \quad (x + yE(x))F(x + yE(x))^{-1} &= xF(x)^{-1} + \alpha(H(x))^{-1}yE(x)F(y)^{-1}. \end{aligned}$$

<sup>(12)</sup>**N.D.E.** The element denoted  $G$  in the original has been denoted  $E$ , since  $G$  already denotes the  $S$ -group under consideration.

<sup>(13)</sup>**N.D.E.** The following has been corrected.

<sup>(14)</sup>**N.D.E.** I.e. one has made the “change of variables”  $x = v, y = wE(v)^{-1}$ , that is,  $v = x, w = yE(x)$ .

<sup>(15)</sup>**N.D.E.**  $yE(x)$  has been corrected to  $x + yE(x)$ .

**C)** Reasoning similarly with associativity on the left, one proves the following assertion:<sup>(16)</sup>

There exists an open subset  $N'$  of  $\mathbf{G}_{a,S}^2$  containing the zero section such that if  $(x, y) \in N'(S')$ , then  $x, y$  and  $x + yF(x)$ <sup>(17)</sup> belong to  $W(S')$ , and

$$(4'') \quad F(x + yF(x)) = F(x)F(y),$$

$$(5'') \quad H(x + yF(x)) = H(x)H(y),$$

$$(6'') \quad (x + yF(x))E(x + yF(x))^{-1} = xE(x)^{-1} + \alpha(H(x))^{-1}yF(x)E(y)^{-1}.$$

We are therefore led to solve the “functional equation” (1'').

**Lemma 2.3.** *Let  $S$  be a scheme,  $W$  an open subset of  $\mathbf{G}_{a,S}$  containing the unit section,  $F : W \rightarrow \mathbf{G}_{m,S}$  an  $S$ -morphism. Suppose that  $F(0) = 1$  and that there exists an open subset  $N$  of  $\mathbf{G}_{a,S}^2$  containing the zero section such that for  $(x, y) \in N(S')$ ,  $x, y$  and  $x + yF(x)$ <sup>(17)</sup> belong to  $W(S')$  and one has:*

$$(\dagger) \quad F(x + yF(x)) = F(x)F(y).$$

(i) *If  $S$  is the spectrum of a field  $k$ , there exists  $a \in k$  such that  $F(x) = 1 + ax$ .*

(ii) *If  $a = F'(0) \in \Gamma(S, \mathcal{O}_S)$  is invertible, then  $F(x) = 1 + ax$ .*

By the hypotheses, we can differentiate the given equation at  $x = 0$  (resp. at  $y = 0$ ), and find that

$$(*) \quad F'(y)(1 + yF'(0)) = F'(0)F(y) \quad \text{for } (0, y) \in N(S'),$$

resp.

$$F'(x)F(x) = F(x)F'(0) \quad \text{for } (x, 0) \in N(S').$$

Since  $F$  takes values in  $\mathbf{G}_m$ , the second relation gives

$$(*') \quad F'(x) = F'(0) \quad \text{for } (x, 0) \in N(S');$$

whence, by the first,

$$F'(0)(1 + yF'(0)) = F'(0)F(y) \quad \text{for } (y, 0), (0, y) \in N(S').$$

If  $a = F'(0)$  is invertible, this gives

$$F(y) = 1 + ay,$$

for  $y$  a section of an open subset of  $W$  containing the unit section, hence schematically dense in  $W$ , which proves (ii). This also proves (i) when  $F'(0) \neq 0$ .

If  $F'(0) = 0$ , then, by (\*'),  $F'(x) = 0$  when  $x$  is “near 0”, hence  $F' = 0$  by schematic density. If  $k$  has characteristic 0,  $F$  is a rational fraction with zero derivative, hence constant and equal to  $F(0) = 1$ .

If  $k$  has characteristic  $p$  and  $F$  is not constant,<sup>(18)</sup> there exist an integer  $n > 0$  and a rational fraction  $F_1 \in k(X)$  such that  $F_1'(X) \neq 0$  and

$$F(X) = F_1(X^{p^n}) = F_1(X)^{p^n}.$$

Substituting in the functional equation, one finds

$$(\dagger_1) \quad F_1(x + yF_1(x)^{p^n}) = F_1(x)F_1(y).$$

Differentiating at  $x = 0$ , one finds

$$(*_1) \quad F_1'(y) = F_1'(0)F_1(y),$$

and differentiating  $(\dagger_1)$  at  $y = 0$ , one obtains

$$(*'_1) \quad F_1'(x)F_1(x)^{p^n} = F_1(x)F_1'(0).$$

<sup>(16)</sup>**N.D.E.** I.e. one writes the equalities resulting from  $p_\alpha(t)p_\alpha(u)p_{-\alpha}(v) = p_\alpha(t+u)p_{-\alpha}(v)$ , and puts  $v = 1$  and  $x = u$ ,  $t = yF(u)$  (i.e.  $y = tF(u)^{-1}$ ).

<sup>(17)</sup>**N.D.E.**  $yF(x)$  has been corrected to  $x + yF(x)$ .

<sup>(18)</sup>**N.D.E.** The following has been corrected.

Since, by hypothesis,  $F'_1(X)$  is an invertible element of  $k(X)$ , one deduces from these two equalities that

$$F_1(X)^{p^n} = 1,$$

hence  $F_1$  is a constant, contradicting the initial hypothesis. This shows that  $F$  is constant and equal to  $1 = F(0)$ .

**D)** Suppose that  $S$  is the spectrum of a field. If  $F'(0) = 0$ , then  $F = 1$ . Formula (5'') then gives  $H(x+y) = H(x)H(y)$ , which shows that  $H$  extends to a morphism of groups  $\mathbf{G}_{a,S} \rightarrow T$  (Exp. XVIII 2.3), which is necessarily constant with value  $e$ . On the other hand, by lemma 2.3, one will also have  $E(x) = 1 + bx$  for some  $b \in k$ . But then (6'') gives, for  $(x, y) \in N(S')$ ,

$$(x+y)E(x+y)^{-1} = xE(x)^{-1} + yE(y)^{-1},$$

hence,<sup>(19)</sup> by Exp. XVIII 2.3 again,  $x \mapsto xE(x)^{-1}$  extends to a morphism of  $k$ -groups  $\mathbf{G}_{a,k} \rightarrow \mathbf{G}_{a,k}$ , hence  $x/(1+bx) = cx$  for some  $c \in k$ , whence  $b = 0$  (and  $c = 1$ ).

This shows that  $F, H, E$  are constant with values  $(1, e, 1)$  in a neighborhood of the unit section, hence everywhere, which by (+) shows that  $U_\alpha$  and  $U_{-\alpha}$  commute, contrary to hypothesis (ii).

If  $S$  is now arbitrary, one has therefore proved that  $F'(0)$  vanishes on no fiber, hence is invertible. The same is clearly true of  $E'(0)$ , which, by lemma 2.3, shows that there exist  $a, b \in \mathbf{G}_m(S)$  such that

$$(\diamond_1) \quad F(x) = 1 + ax, \quad E(x) = 1 + bx, \quad \text{for } x \in W(S').$$

**E)** The rest is now easy. Substituting the preceding results in (3''), one finds

$$y\alpha(H(x))(1+ay) = y(1+ax+ay(1+bx))(1+ax).$$

This formula is valid for every section  $(x, y)$  of  $N$ . But since  $\mathbf{G}_{a,S} \times_S \mathbf{G}_{m,S}$  is schematically dense in  $\mathbf{G}_{a,S}^2$ , one deduces

$$(1+ay)\alpha(H(x)) = (1+ax+ay(1+bx))(1+ax).$$

Putting  $y = 0$  gives  $\alpha(H(x)) = (1+ax)^2$ . Substituting this in the preceding equality, one finds<sup>(20)</sup>

$$a^2xy = abxy.$$

Since  $\mathbf{G}_{m,S}$  is schematically dense in  $\mathbf{G}_{a,S}$ , one deduces  $a^2 = ab$ , whence, since  $a$  is invertible,

$$(\diamond_2) \quad a = b \quad \text{and} \quad \alpha(H(x)) = (1+ax)^2.$$

Since  $a$  is invertible,  $x \mapsto 1+ax$  is an automorphism of  $\mathbf{G}_{a,S}$ ; one can therefore find an open subset  $W'$  of  $\mathbf{G}_{a,S}$  containing the section 1 and a morphism

$$P : W' \rightarrow T$$

such that  $P(1+ax) = H(x)$ .<sup>(21)</sup>

Substituting in relation (2'), one finds at once, for  $(x, y) \in N(S')$ ,

$$P(1+ax+ay) = P\left(\frac{1+ax+ay}{1+ax}\right)P(1+ax),$$

which proves that there exists an open neighborhood of 1 in  $\mathbf{G}_{m,S}$  such that for  $x$  and  $y$  in this neighborhood one has  $P(x)P(y) = P(xy)$ . By Exp. XVIII 2.3, there exists a morphism of groups

$$(\diamond_3) \quad \alpha^* : \mathbf{G}_{m,S} \rightarrow T$$

<sup>(19)</sup>**N.D.E.** The following sentence has been added. One can also see by a direct calculation that the preceding equality implies  $0 = xyb(2 + (x+y)b)$ , whence  $0 = b(2 + (x+y)b)$ , and finally  $b = 0$ .

<sup>(20)</sup>**N.D.E.** In the preceding three equalities, the original has been corrected by replacing the factor  $(1+bx)$  on the right by  $1+ax$ . Substituting the third equality in the second and taking account of the invertibility of  $1+ax$ , one indeed obtains  $a^2xy = abxy$ .

<sup>(21)</sup>**N.D.E.** I.e. one has made the change of variables  $x' = 1+ax$ , that is,  $x = (x' - 1)/a$ .

extending  $P$ . Since  $\alpha(H(x)) = (1 + ax)^2$  in a neighborhood of the zero section, one has  $\alpha(\alpha^*(z)) = z^2$  in a neighborhood of the section 1, hence

$$(\diamond_4) \quad \alpha \circ \alpha^* = 2.$$

**F)** <sup>(22)</sup>Collecting results (+) and  $(\diamond_1 - \diamond_4)$ , one sees that there exist  $a \in \mathbf{G}_m(S)$  and  $\alpha^* \in \text{Hom}_{S\text{-gr.}}(\mathbf{G}_{m,S}, T)$  such that  $\alpha \circ \alpha^* = 2$ , and such that, if  $(u, v) \in V(S')$  and  $uv \in W(S')$ , then  $1 + auv$  is invertible and

$$p_\alpha(u)p_{-\alpha}(v) = p_{-\alpha}\left(\frac{v}{1 + auv}\right)\alpha^*(1 + auv)p_\alpha\left(\frac{u}{1 + auv}\right).$$

Consider the open subset  $V'$  of  $\mathbf{G}_{a,S}^2$  defined by “ $1 + auv$  invertible”, i.e.  $V' = (\mathbf{G}_{a,S}^2)_f$ , where  $f(u, v) = 1 + auv$ . Both sides of the preceding formula define morphisms from  $V'$  to  $G$  which coincide in a neighborhood of the zero section, hence coincide in  $V'$ . The preceding formula is therefore valid for every section  $(u, v)$  of  $V'$ . In particular, it follows that  $V' \subset U$ , where  $U$  is the open subset introduced at the beginning of A).

Let us prove that  $U = V'$ . Returning to the notations of A), one has a morphism

$$A : U \longrightarrow \mathbf{G}_{a,S}$$

which, on  $V'$ , is defined by  $A(u, v) = v(1 + auv)^{-1}$ . To show that  $U = V'$ , which is a set-theoretic question, one reduces to the case where  $S$  is the spectrum of a field  $k$ , hence to the following obvious assertion: the domain of definition of the rational map  $\mathbf{G}_{a,k}^2 \longrightarrow \mathbf{G}_{a,k}$  defined by the rational fraction  $Y/(1 + aXY)$  is the open subset defined by the function  $1 + aXY$ .

**G)** One has therefore proved the existence of  $a$  and  $\alpha^*$ , as well as the two supplementary properties announced. It remains to prove uniqueness. Let therefore  $a'$  and  $\alpha^{*'} also satisfy the required conditions. If  $u, v \in \mathbf{G}_a(S')^2$ , one has at once:$

$$1 + auv \text{ invertible} \Rightarrow 1 + a'uv \text{ invertible and } \frac{v}{1 + auv} = \frac{v}{1 + a'uv};$$

one therefore has for every section  $u$  of  $\mathbf{G}_a(S')$

$$1 + au \text{ invertible} \Rightarrow 1 + au = 1 + a'u,$$

which proves at once that  $a = a'$ .

With the same notations, one then has

$$1 + au \text{ invertible} \Rightarrow \alpha^*(1 + au) = \alpha^{*'}(1 + au),$$

hence also  $\alpha^* = \alpha^{*'}$ .

**Corollary 2.4.** *Let  $\exp(Y)t\exp(X)$  and  $\exp(Y')t'\exp(X')$  be two elements of  $\Omega(S')$ . Then their product is in  $\Omega(S')$  if and only if  $u = 1 + \langle X, Y' \rangle$  is invertible, and one then has*

$$(F') \quad \begin{aligned} \exp(Y)t\exp(X) \cdot \exp(Y')t'\exp(X') &= \exp(Y + u^{-1}\alpha(t)^{-1}Y') \\ &\quad \cdot tt'\alpha^*(u) \cdot \exp(u^{-1}\alpha(t')^{-1}X + X'). \end{aligned}$$

**Remark 2.5.** One can also write formula (F) of theorem 2.1 without using the morphisms  $\exp$ . Indeed, transporting by these morphisms the duality  $\mathfrak{g}^\alpha \otimes \mathfrak{g}^{-\alpha} \longrightarrow \mathcal{O}_S$ , one obtains a canonical pairing of vector bundles:

$$U_\alpha \times_S U_{-\alpha} \longrightarrow \mathbf{G}_{a,S},$$

which we shall again denote by  $(x, y) \longmapsto \langle x, y \rangle$ . One therefore has

$$\langle \exp X, \exp Y \rangle = \langle X, Y \rangle.$$

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<sup>(22)</sup>**N.D.E.** The following has been slightly modified, since an open subset  $V$  has already been introduced in A).

If  $x \in U_\alpha(S')$ ,  $y \in U_{-\alpha}(S')$  and  $1 + \langle x, y \rangle \in \mathbf{G}_m(S')$ , one has

$$(F) \quad x \cdot y = y^{(1+\langle x, y \rangle)^{-1}} \cdot \alpha^*(1 + \langle x, y \rangle) \cdot x^{(1+\langle x, y \rangle)^{-1}}.$$

**Corollary 2.6.** *The pairing*

$$W(\mathfrak{g}^\alpha) \times_S W(\mathfrak{g}^{-\alpha}) \longrightarrow \mathbf{G}_{a,S}$$

*defines a pairing of principal bundles under  $\mathbf{G}_{m,S}$*

$$W(\mathfrak{g}^\alpha)^\times \times_S W(\mathfrak{g}^{-\alpha})^\times \longrightarrow \mathbf{G}_{m,S}.$$

*This pairing will be denoted  $(X, Y) \mapsto \langle X, Y \rangle$ , or more simply  $(X, Y) \mapsto XY$ .*

*For every section  $X \in \Gamma(S, \mathfrak{g}^\alpha)^\times$ , there therefore exists a unique section  $X^{-1}$  of  $\Gamma(S, \mathfrak{g}^{-\alpha})^\times$  such that  $XX^{-1} = 1$ . One has  $(zX)^{-1} = z^{-1}X^{-1}$ . The morphism*

$$s : W(\mathfrak{g}^\alpha)^\times \longrightarrow W(\mathfrak{g}^{-\alpha})^\times$$

*thus defined is therefore an isomorphism of schemes, compatible with the isomorphism  $s : z \mapsto z^{-1}$  on the operator groups.*

**Definition 2.6.1.** One will say that  $X$  and  $s(X) = X^{-1}$  are *paired*.

Apply corollary 2.4 to  $Y = 0 = X'$  and  $Y' = aX^{-1}$ ,  $a \in \mathcal{O}_S(S)$ . Then  $u = 1 + a$  and  $u^{-1}Y' = u^{-1}(u - 1)X^{-1} = (1 - u^{-1})X^{-1}$ , whence:

**Corollary 2.7.** *Let  $X \in \Gamma(S, \mathfrak{g}^\alpha)^\times$  and  $u \in \Gamma(S, \mathcal{O}_S)^\times$ . One has*

$$\alpha^*(u) = \exp((u^{-1} - 1)X^{-1}) \exp(X) \exp((u - 1)X^{-1}) \exp(-u^{-1}X).$$

**Definition 2.8.** The morphism  $\alpha^*$  is called the *coroot* associated with the root  $\alpha$ .

**Remark 2.9.** If  $(G, T, \alpha)$  is an elementary  $S$ -system,  $(G, T, -\alpha)$  is also one. One therefore has by theorem 2.1 a duality between  $\mathfrak{g}^{-\alpha}$  and  $\mathfrak{g}^\alpha$ , and a coroot  $(-\alpha)^*$ . Taking the inverse of formula (F), one proves at once

$$\langle X, Y \rangle = \langle Y, X \rangle, \quad (-\alpha)^* = -\alpha^*.$$

Let us now pass to the Lie algebra of  $G$ . The root  $\alpha$  and the coroot  $\alpha^*$  define the linear forms

$$\mathcal{O}_S \xrightarrow{\bar{\alpha}^*} \mathfrak{t} \xrightarrow{\bar{\alpha}} \mathcal{O}_S.$$

One puts  $H_\alpha = \bar{\alpha}^*(1)$ . One calls  $\bar{\alpha}$  the *infinitesimal root* associated with  $\alpha$ , and  $H_\alpha$  the corresponding *infinitesimal coroot*.

**Lemma 2.10.** *Let  $S' \longrightarrow S$ , and  $X, X' \in W(\mathfrak{g}^\alpha)(S')$ ,  $H \in W(\mathfrak{t})(S')$ ,  $Y, Y' \in W(\mathfrak{g}^{-\alpha})(S')$ ,  $t \in T(S')$ . One has*

- (1)  $\text{Ad}(t)H = H, \quad \text{Ad}(t)X = \alpha(t)X, \quad \text{Ad}(t)Y = \alpha(t)^{-1}Y.$
- (2)  $\begin{cases} \text{Ad}(\exp(X))H = H - \bar{\alpha}(H)X, & \text{Ad}(\exp(X))X' = X', \\ \text{Ad}(\exp(X))Y = Y + \langle X, Y \rangle H_\alpha - \langle X, Y \rangle X. \end{cases}$
- (2')  $\begin{cases} \text{Ad}(\exp(Y))H = H + \bar{\alpha}(H)Y, & \text{Ad}(\exp(Y))Y' = Y', \\ \text{Ad}(\exp(Y))X = X + \langle X, Y \rangle H_{-\alpha} - \langle X, Y \rangle Y. \end{cases}$
- (3)  $[H, X] = \bar{\alpha}(H)X, \quad [H, Y] = -\bar{\alpha}(H)Y, \quad [X, Y] = \langle X, Y \rangle H_\alpha.$
- (4)  $H_{-\alpha} = -H_\alpha.$
- (5)  $\bar{\alpha}(H_\alpha) = 2.$

The proof of these various formulas is either trivial or an immediate consequence of formula (F) of 2.1.

**Corollary 2.11.** *Suppose that  $H_\alpha$  is nonzero on every fiber (which is in particular the case if 2 is invertible on  $S$ , by (5)). Then  $X_\alpha \in \Gamma(S, \mathfrak{g}^\alpha)^\times$  and  $X_{-\alpha} \in \Gamma(S, \mathfrak{g}^{-\alpha})^\times$  are paired if and only if  $[X_\alpha, X_{-\alpha}] = H_\alpha$ .*

**2.12.** — Let  $(G, T, \alpha)$  be an elementary  $S$ -system. We know (1.19) that the center of  $G$  is  $\text{Centr}(G) = \text{Ker}(\alpha)$ , a group of multiplicative type and of finite type. If  $Q$  is a subgroup of multiplicative type of  $\text{Centr}(G)$ , the quotient  $G/Q$  is affine over  $S$  (Exp. IX 2.5), smooth over  $S$  (Exp. VI<sub>B</sub> 9.2), with connected reductive fibers of semisimple rank 1 (Exp. XIX 1.8).

Put  $G' = G/Q$ ; it is a reductive  $S$ -group of semisimple rank 1;  $T' = T/Q$  is a maximal torus of it. The open subset  $U_{-\alpha} \cdot T \cdot U_\alpha$  of  $G$  is stable under  $Q$ , and one sees at once that the quotient is isomorphic to  $U_{-\alpha} \times_S (T/Q) \times_S U_\alpha$ . If  $\alpha'$  denotes the character of  $T'$  induced by  $\alpha$ , it follows that the derived morphism of the canonical morphism  $G \rightarrow G'$  induces isomorphisms

$$\mathfrak{g}^\alpha \xrightarrow{\sim} \mathfrak{g}'^{\alpha'} \quad \text{and} \quad \mathfrak{g}^{-\alpha} \xrightarrow{\sim} \mathfrak{g}'^{-\alpha'}.$$

In particular,  $\alpha'$  is a root of  $G'$  with respect to  $T'$ . Thus, denoting by  $\alpha/Q$  the character  $T/Q \rightarrow \mathbf{G}_{m,S}$  induced by  $\alpha$ , one has:

**Lemma 2.13.** *If  $Q$  is a subgroup of multiplicative type of  $\text{Ker}(\alpha)$ , then*

$$(G/Q, T/Q, \alpha/Q)$$

*is an elementary system.*

**Lemma 2.14.** *Under the preceding conditions, the following diagrams commute:*

$$\begin{array}{ccccc} W(\mathfrak{g}^\alpha) & \xrightarrow{\text{exp}} & G & \xleftarrow{\text{exp}} & W(\mathfrak{g}^{-\alpha}) \\ \text{can} \downarrow \simeq & & \downarrow \text{can} & & \simeq \downarrow \text{can} \\ W(\mathfrak{g}'^{\alpha'}) & \xrightarrow{\text{exp}} & G' & \xleftarrow{\text{exp}} & W(\mathfrak{g}'^{-\alpha'}) \\ \\ \mathfrak{g}^\alpha \otimes \mathfrak{g}^{-\alpha} & \xrightarrow{\sim} & \mathcal{O}_S & & \\ \text{can} \downarrow \simeq & & \downarrow \text{id} & & \\ \mathfrak{g}'^{\alpha'} \otimes \mathfrak{g}'^{-\alpha'} & \xrightarrow{\sim} & \mathcal{O}_S & & \\ \\ \begin{array}{ccc} & T & \\ \alpha^* \nearrow & \downarrow \text{can} & \searrow \alpha \\ \mathbf{G}_{m,S} & & \mathbf{G}_{m,S} \\ \alpha'^* \searrow & \downarrow & \nearrow \alpha' \\ & T' & \end{array} & & \end{array}.$$

### 3. The Weyl group

**Notation 3.0.** <sup>(23)</sup>If  $(G, T, \alpha)$  is an elementary  $S$ -system, one puts

$$N = \underline{\text{Norm}}_G(T), \quad W = \underline{\text{Norm}}_G(T)/T$$

(cf. Exp. XIX 6.3);  $N$  is a closed subgroup of  $G$ , smooth over  $S$ . One denotes by  $N^\times = N - T$  the open subscheme of  $N$  induced on the complement of  $T$ .<sup>(24)</sup> Let  $R$  be the (unique) maximal torus of  $\text{Ker}(\alpha)$ , and  $T'$  the image of  $\alpha^* : \mathbf{G}_{m,S} \rightarrow T$ , which is a one-dimensional subtorus of  $T$ .

<sup>(23)</sup>**N.D.E.** The numbering 3.0 has been added for subsequent references.

<sup>(24)</sup>**N.D.E.**  $Q$  has been replaced by the more suggestive notation  $N^\times$ .

The morphism

$$T' \times_S R \longrightarrow T$$

induced by the product in  $T$  is surjective (hence faithfully flat); indeed, one reduces to checking this on geometric fibers, and it follows at once from the formula  $\alpha \circ \alpha^* = 2$ .

**Theorem 3.1.** *With the preceding notations:*

- (i)  $W$  is isomorphic to the constant group  $(\mathbf{Z}/2\mathbf{Z})_S$ .
- (ii)  $N^\times$  is a locally trivial principal homogeneous bundle under  $T$ , on the left by the law  $(t, q) \mapsto tq$  (resp. on the right by the law  $(q, t) \mapsto qt$ ).
- (iii) One has the formula

$$\text{int}(w)t = t \cdot \alpha^*(\alpha(t)^{-1})$$

for  $w \in N^\times(S')$ ,  $t \in T(S')$ ,  $S' \longrightarrow S$ . In the decomposition  $T_{S'} = T'_{S'} \cdot R_{S'}$ ,  $\text{int}(w)$  induces the identity on  $R_{S'}$  and reflection on  $T'_{S'}$ . One has the relations

$$\alpha \circ \text{int}(w) = \alpha^{-1}, \quad \text{int}(w) \circ \alpha^* = (\alpha^*)^{-1}.$$

- (iv) For  $X \in W(\mathfrak{g}^\alpha)^\times(S')$ ,  $S' \longrightarrow S$ , put

$$w_\alpha(X) = \exp(X) \exp(-X^{-1}) \exp(X).$$

Then  $w_\alpha(X) \in N^\times(S')$ , and the morphism  $w_\alpha : W(\mathfrak{g}^\alpha)^\times \longrightarrow N^\times$  thus defined satisfies

$$w_\alpha(zX) = \alpha^*(z)w_\alpha(X) = w_\alpha(X)\alpha^*(z)^{-1},$$

for  $z \in \mathbf{G}_m(S')$ ,  $X \in W(\mathfrak{g}^\alpha)^\times(S')$ ,  $S' \longrightarrow S$ .

- (v) <sup>(25)</sup>For  $X, Y \in W(\mathfrak{g}^\alpha)^\times(S')$ ,  $S' \longrightarrow S$ , with the notations of 2.6, one has

$$w_\alpha(X)w_\alpha(Y) = \alpha^\vee(-XY^{-1}).$$

In particular,

$$w_\alpha(X)^2 = \alpha^*(-1) \in {}_2T(S) \cap \underline{\text{Centr}}(G)(S),$$

$$w_\alpha(X)^{-1} = w_\alpha(-X) = \alpha^*(-1)w_\alpha(X).$$

- (vi) If one similarly defines, for  $Y \in W(\mathfrak{g}^{-\alpha})^\times(S')$ ,

$$w_{-\alpha}(Y) = \exp(Y) \exp(-Y^{-1}) \exp(Y),$$

one has (in addition to formulas analogous to the preceding ones)

$$w_{-\alpha}(X^{-1}) = w_\alpha(X)^{-1} = w_\alpha(-X),$$

$$w_\alpha(X)w_{-\alpha}(Y) = \alpha^*(XY).$$

*Proof.* (i) was already seen in Exp. XIX 2.4; it follows at once that  $N^\times$  is indeed a principal homogeneous bundle under  $T$  for the laws defined in (ii); its local triviality (for the Zariski topology) follows, in particular, from (iv).

Let us prove (iii); if  $w \in N^\times(S)$ , clearly  $\alpha \circ \text{int}(w)$  is a root of  $G$  with respect to  $T$ , hence is locally equal to  $\alpha$  or  $-\alpha$ ; since on each fiber it is  $-\alpha$  (*Bible*, 12-05, proof of the cor. to prop. 1), one has  $\alpha \circ \text{int}(w) = -\alpha$ . By transport of structure, one deduces

$$-\alpha^* = \text{int}(w)^{-1} \circ \alpha^* = \text{int}(w) \circ \alpha^*,$$

since  $\text{int}(w)^2 = \text{int}(w^2)$  and  $w^2$  is a section of  $T$ . Thus  $\text{int}(w)$  induces reflection on  $T'$ ; since  $R$  is central,  $\text{int}(w)$  induces the identity on  $R$ . The formula of (iii) defines a morphism  $T \longrightarrow T$  with the same properties, hence coincides with  $\text{int}(w)$ .

Let us prove (iv). One has successively

$$\begin{aligned} w_\alpha(X)tw_\alpha(X)^{-1} &= \exp(X) \exp(-X^{-1}) \exp(X)t \exp(-X) \exp(X^{-1}) \exp(-X) \\ &= \exp(X) \exp(-X^{-1}) \exp(X - \alpha(t)X) \exp(\alpha(t)^{-1}X^{-1}) \exp(-\alpha(t)X)t. \end{aligned}$$

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<sup>(25)</sup>**N.D.E.** The first formula of the original has been corrected.

Applying formula (F), one has

$$\begin{aligned} \exp(-X^{-1}) \exp((1 - \alpha(t))X) &= \exp((\alpha(t)^{-1} - 1)X) \alpha^*(\alpha(t)^{-1}) \\ &\quad \cdot \exp(-\alpha(t)^{-1}X^{-1}). \end{aligned}$$

Substituting in the preceding relation, one finds

$$\begin{aligned} \text{int}(w_\alpha(X))t &= \exp(\alpha(t)^{-1}X) \alpha^*(\alpha(t)^{-1}) \exp(-\alpha(t)X)t \\ &= \exp(aX) \alpha^*(\alpha(t)^{-1})t, \end{aligned}$$

where

$$a = \alpha(t)^{-1} - (\alpha \circ \alpha^*)(\alpha(t)^{-1})\alpha(t),$$

but  $\alpha \circ \alpha^* = 2$ , which gives at once  $a = 0$  and  $w_\alpha(X) \in N^\times(S')$ .

Let us now prove the second assertion of (iv). One has<sup>(26)</sup>

$$\begin{aligned} \alpha^*(z)w_\alpha(X) &= \exp(z^2X) \exp(-z^{-2}X^{-1}) \exp(z^2X) \alpha^*(z) \\ &= \exp(zX) \exp((z^2 - z)X) \exp(-z^{-2}X^{-1}) \exp(z^2X) \alpha^*(z) \\ &= \exp(zX) \exp(-z^{-1}X^{-1}) \alpha^*(z)^{-1} \exp((z^3 - z^2)X) \exp(z^2X) \alpha^*(z) \\ &= \exp(zX) \exp(-z^{-1}X^{-1}) \exp(zX) = w_\alpha(zX). \end{aligned}$$

Let us prove (v). By the preceding result, the first formula of (v) follows at once from the second; let us prove the latter:

$$\begin{aligned} w_\alpha(X)^2 &= \exp(X) \exp(-X^{-1}) \exp(2X) \exp(-X^{-1}) \exp(X) \\ &= \exp(X) \exp(-X^{-1}) \exp(X^{-1}) \alpha^*(-1) \exp(-2X) \exp(X) \\ &= \exp(X) \alpha^*(-1) \exp(-X) = \alpha^*(-1), \end{aligned}$$

since  $\alpha(\alpha^*(-1)) = (-1)^2 = 1$ , which proves that  $\alpha^*(-1) \in \text{Centr}(G)(S)$ .

Finally, let us prove (vi). The first assertion is a particular case of the second; let us prove the latter. Both sides of this formula define morphisms from  $W(\mathfrak{g}^\alpha)^\times \times_S W(\mathfrak{g}^{-\alpha})^\times$  to  $G$ . To prove that they coincide, it suffices to do so on an open subset nonempty on each fiber (Exp. XVIII 1.4); it therefore suffices to check the relation when  $1 + XY$  is invertible. One then has successively:

$$\begin{aligned} w_\alpha(X)w_{-\alpha}(Y) &= \exp(X) \exp(-X^{-1}) \exp(X) \exp(Y) \exp(-Y^{-1}) \exp(Y) \\ &= \exp(X) \exp(-X^{-1}) \exp\left(\frac{Y}{1+XY}\right) \alpha^*(1+XY) \\ &\quad \exp\left(\frac{X}{1+XY}\right) \exp(-Y^{-1}) \exp(Y) \\ &= \exp(X) \exp\left(\frac{-X^{-1}}{1+XY}\right) \alpha^*(1+XY) \exp\left(\frac{-Y^{-1}}{1+XY}\right) \exp(Y) \\ &= \exp(-X^{-2}Y^{-1}) \alpha^*\left(\frac{XY}{1+XY}\right) \exp(X+Y^{-1}) \\ &\quad \alpha^*(1+XY) \exp\left(\frac{-Y^{-1}}{1+XY}\right) \exp(Y) \\ &= \exp(-X^{-2}Y^{-1}) \alpha^*(XY) \exp\left(\frac{Y^{-1}+X}{(1+XY)^2}\right) \exp\left(\frac{-Y^{-1}}{1+XY}\right) \exp(Y) \\ &= \alpha^*(XY) \exp(-Y) \exp(Y) = \alpha^*(XY). \end{aligned}$$

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<sup>(26)</sup>**N.D.E.** The first equality follows from 1.5 (i), which, combined with  $\alpha \circ \alpha^* = 2$ , gives the formulas

$$\begin{aligned} \alpha^*(z) \exp(X) \alpha^*(z)^{-1} &= \exp(z^2X), \\ \alpha^*(z) \exp(X^{-1}) \alpha^*(z)^{-1} &= \exp(z^{-2}X), \end{aligned}$$

the third equality follows from formula (F), and the fourth from (†), again. Finally, a similar calculation shows that  $w_\alpha(X) \alpha^*(z^{-1}) = w_\alpha(zX)$ .

**Corollary 3.2.** *Let  $n \in \mathbf{Z}$ ,  $n \neq 0$ . For every  $w \in G(S)$ , the following conditions are equivalent:*

- (i)  $w \in N^\times(S)$ ,
- (ii) *one has  $\text{int}(w) \circ n\alpha^* = -n\alpha^*$  (recall that  $(n\alpha^*)(z) = \alpha^*(z)^n$ ).*

One has (i)  $\Rightarrow$  (ii) (assertion (iii) of theorem 3.1); conversely, one may suppose that  $N^\times$  has a section, and one is reduced to proving:

**Lemma 3.3.** *One has  $\underline{\text{Centr}}_G(n\alpha^*) = T$  for  $n \neq 0$ .*

Indeed, the image  $T'$  of  $n\alpha^*$  is a subtorus of  $G$ . It follows (Exp. XIX 2.8) that  $\underline{\text{Centr}}_G(n\alpha^*)$  is a reductive subgroup of  $G$ , containing  $T$ . Since on each fiber one has  $\underline{\text{Centr}}_{G_{\bar{s}}}(n\alpha_{\bar{s}}^*) \neq G_{\bar{s}}$ , one has  $\underline{\text{Centr}}_{G_{\bar{s}}}(n\alpha_{\bar{s}}^*) = T_{\bar{s}}$  (Exp. XIX 1.6.3<sup>(27)</sup>), hence  $\underline{\text{Centr}}_G(n\alpha^*) = T$ , since these are smooth subgroups of  $G$ .

**Remark 3.4.** The construction of  $w_\alpha$  and the fact that  $w_\alpha(X)$  normalizes  $T$  rely only on formula (F). In particular, if  $G$  is an  $S$ -group satisfying the conditions of 2.2,  $\underline{\text{Norm}}_G(T)$  is different from  $T$  on each fiber. It follows that if  $G$  is an affine  $S$ -group with connected fibers satisfying the conditions of 2.2, it is reductive of semisimple rank 1. Indeed, it is smooth in a neighborhood of the unit section, hence smooth, and one can apply the criterion of Exp. XIX 1.11.

**3.5.** — Before stating the following theorem, let us make a few remarks. As usual, we identify  $\mathfrak{g}^{-\alpha}$  with  $(\mathfrak{g}^\alpha)^{\otimes -1}$ . Similarly, we shall identify  $\underline{\text{Hom}}_{\mathcal{O}_S}(\mathfrak{g}^{-\alpha}, \mathfrak{g}^\alpha)$  with  $(\mathfrak{g}^\alpha)^{\otimes 2}$ , and hence

$$\underline{\text{Isom}}_{\mathcal{O}_S\text{-mod.}}(\mathbf{W}(\mathfrak{g}^{-\alpha}), \mathbf{W}(\mathfrak{g}^\alpha)) \simeq \mathbf{W}((\mathfrak{g}^\alpha)^{\otimes 2})^\times.$$

If  $w \in N^\times(S)$ ,  $\text{Ad}(w)$  interchanges  $\mathfrak{g}^\alpha$  and  $\mathfrak{g}^{-\alpha}$  (3.1, (iii)), hence defines an isomorphism

$$a_\alpha(w) : \mathfrak{g}^{-\alpha} \xrightarrow{\sim} \mathfrak{g}^\alpha,$$

which we shall therefore identify with a section  $a_\alpha(w) \in \Gamma(S, (\mathfrak{g}^\alpha)^{\otimes 2})^\times$ . This construction is compatible with base change and therefore defines a morphism

$$a_\alpha : N^\times \longrightarrow \mathbf{W}((\mathfrak{g}^\alpha)^{\otimes 2})^\times,$$

such that  $a_\alpha(w)Y = \text{Ad}(w)Y$  for all  $w \in N^\times(S')$ ,  $Y \in \Gamma(S', \mathfrak{g}^{-\alpha})^\times$ ,  $S' \longrightarrow S$ .

**Theorem 3.6.**

- (i) *One has*

$$\text{int}(w) \exp(Y) = \exp(a_\alpha(w)Y)$$

*for every  $S' \longrightarrow S$  and all  $w \in N^\times(S')$ ,  $Y \in \mathbf{W}(\mathfrak{g}^{-\alpha})(S')$ .*

- (ii) *One has*

$$a_\alpha(tw) = \alpha(t)a_\alpha(w), \quad a_\alpha(wt) = \alpha(t)^{-1}a_\alpha(w).$$

- (iii) *If one similarly defines  $a_{-\alpha} : N^\times \longrightarrow \mathbf{W}((\mathfrak{g}^{-\alpha})^{\otimes 2})^\times$ , one has*

$$a_{-\alpha}(w) = a_\alpha(w)^{-1}.$$

(28)

- (iv) *For every  $X \in \mathbf{W}(\mathfrak{g}^\alpha)^\times(S')$ ,  $S' \longrightarrow S$ , one has*

$$a_\alpha(w_\alpha(X)) = -X^2.$$

<sup>(27)</sup>**N.D.E.** The hypothesis  $\underline{\text{Centr}}_{G_{\bar{s}}}(n\alpha_{\bar{s}}^*) \neq G_{\bar{s}}$  implies that  $\dim \underline{\text{Centr}}_{G_{\bar{s}}}(n\alpha_{\bar{s}}^*) - \dim T_{\bar{s}} < 2$ , whereas this difference is even, by loc. cit.

<sup>(28)</sup>**N.D.E.** I.e.  $a_{-\alpha}(w)$  and  $a_{-\alpha}(w)$  are paired, cf. 2.6.1.

Assertion (i) is trivial by the characterization of the morphisms  $\exp$  given in 1.5. Assertion (ii) is immediate, as is (iii). Let us prove (iv): let  $X \in \Gamma(S', \mathfrak{g}^\alpha)^\times$ ,  $Z \in \Gamma(S', \mathfrak{g}^\alpha)$ ; by definition, one has<sup>(29)</sup>

$$\begin{aligned} a_\alpha(w_\alpha(X))^{-1}(Z) &= \text{Ad}(w_\alpha(X))(Z) \\ &= \text{Ad}(\exp(X)) \text{Ad}(\exp(-X^{-1})) \text{Ad}(\exp(X))(Z). \end{aligned}$$

Applying formulas (2') and (2) of lemma 2.10, as well as the equalities  $H_{-\alpha} = -H_\alpha$ ,  $\bar{\alpha}(H_\alpha) = 2$  (loc. cit. (4) and (5)) and  $\langle X, X^{-1} \rangle = 1$  (2.6), one obtains that the right-hand term equals, successively:

$$\begin{aligned} \text{Ad}(\exp(X)) \text{Ad}(\exp(-X^{-1}))(Z) &= \text{Ad}(\exp(X))(Z + \langle X^{-1}, Z \rangle (H_\alpha - X^{-1})) \\ &= Z + \langle X^{-1}, Z \rangle (H_\alpha - 2X - X^{-1} - H_\alpha + X) \\ &= Z - \langle X^{-1}, Z \rangle X - \langle X^{-1}, Z \rangle X^{-1}. \end{aligned}$$

But  $Z = \langle X^{-1}, Z \rangle X$  and  $\langle X^{-1}, Z \rangle X^{-1} = X^{-2}Z$ , hence this shows that  $a_\alpha(w_\alpha(X))^{-1} = -X^{-2}$ , whence  $a_\alpha(w_\alpha(X)) = -X^2$ .

**Corollary 3.7.** *In particular, one has*

$$\text{int}(w_\alpha(X)) \exp(X) = \exp(-X^{-1}),$$

whence (by the definition of  $w_\alpha(X)$ ):

$$w_\alpha(X) \exp(X) w_\alpha(X)^{-1} = \exp(-X) w_\alpha(X) \exp(-X),$$

i.e. by an immediate calculation,

$$(w_\alpha(X) \exp(X))^3 = e.$$

**Corollary 3.8.** *Let  $X \in \Gamma(S, \mathfrak{g}^\alpha)^\times$  and  $n \in \mathbf{Z}$ ,  $n \neq 0$ . Then  $w_\alpha(X)$  is the unique section  $w \in G(S)$  satisfying*

- (i)  $\text{int}(w) \circ n\alpha^* = -n\alpha^*$ .
- (ii)  $(w \exp(X))^3 = e$ .

One knows that  $w_\alpha(X)$  indeed satisfies these conditions. Conversely, let  $w \in G(S)$  satisfy (i) and (ii). By 3.2 and 3.1 (ii), one knows that there exists  $t \in T(S)$  such that  $w = w_\alpha(X)t$ . Put  $u = \exp(X)$ . One then has

$$wuw^{-1} = w_\alpha(X)t \exp(X)t^{-1}w_\alpha(X)^{-1} = \exp(-\alpha(t)X^{-1}),$$

and, on the other hand,

$$\begin{aligned} u^{-1}wu^{-1} &= \exp(-X)w_\alpha(X)t \exp(-X) \\ &= \exp(-X)w_\alpha(X) \exp(-X) \exp(X - \alpha(t)X)t \\ &= \exp(-X^{-1}) \exp(X - \alpha(t)X)t = \exp(-X^{-1})t \exp(H). \end{aligned}$$

Now  $(wu)^3 = e \iff wuw^{-1} = u^{-1}wu^{-1}$ ; comparing the two decompositions of this element on  $U_{-\alpha} \cdot T \cdot U_\alpha$ , one obtains  $t = e$ .

**Remark 3.9.** One can summarize a number of the results of this section by the following diagram of principal homogeneous bundles (on the left):

$$\begin{array}{ccc} \mathbf{W}(\mathfrak{g}^\alpha)^\times & \xrightarrow{w_\alpha} & N^\times \xrightarrow{a_\alpha} \mathbf{W}((\mathfrak{g}^\alpha)^{\otimes 2})^\times \\ & & \\ \mathbf{G}_{\mathbf{m},S} & \xrightarrow{\alpha^*} & T \xrightarrow{\alpha} \mathbf{G}_{\mathbf{m},S}. \end{array}$$

<sup>(29)</sup>**N.D.E.** The following has been expanded.

Note that  $a_\alpha$  is faithfully flat (since  $\alpha$  is) and that  $w_\alpha$  is a monomorphism if and only if  $\alpha^*$  is a monomorphism. We leave it to the reader to write the corresponding diagrams for the structures of principal bundles on the right, as well as diagrams of the same kind for the root  $-\alpha$ , and to study the relations between these various diagrams.

**Lemma 3.10.** *Let  $S$  be a scheme,  $q$  an integer  $> 0$  such that  $x \mapsto x^q$  defines an endomorphism of  $\mathbf{G}_{a,S}$ ,  $(G, T, \alpha)$  and  $(G', T', \alpha')$  two elementary  $S$ -systems,  $f : G \rightarrow G'$  a morphism of  $S$ -groups. Let*

$$h : (\mathfrak{g}^\alpha)^{\otimes q} \xrightarrow{\sim} \mathfrak{g}'^{\alpha'}$$

*be an isomorphism of  $\mathcal{O}_S$ -modules, and*

$$h^\vee : (\mathfrak{g}^{-\alpha})^{\otimes q} \xrightarrow{\sim} \mathfrak{g}'^{-\alpha'}$$

*the contragredient isomorphism. For every  $S' \rightarrow S$  and every  $X \in W(\mathfrak{g}^\alpha)(S')$ , suppose that:*

$$f(\exp(X)) = \exp(h(X^q)).$$

*Then the following conditions are equivalent:*

- (i)  $f(\alpha^*(z)) = \alpha'^*(z)^q$ .
- (ii)  $f(w_\alpha(Z)) = w_{\alpha'}(h(Z^q))$ .
- (iii)  $f(\exp(Y)) = \exp(h^\vee(Y^q))$ .

*(Each condition is to be read: for every  $S' \rightarrow S$  and every  $z \in \mathbf{G}_m(S')$ ,  $Z \in W(\mathfrak{g}^\alpha)^\times(S')$ ,  $Y \in W(\mathfrak{g}^{-\alpha})(S')$ , one has ...)*

Indeed, (i)  $\Rightarrow$  (ii) by 3.8, (ii)  $\Rightarrow$  (iii) by 3.7, (iii)  $\Rightarrow$  (i) by 2.7.

**Proposition 3.11.** *Let  $S$  be a scheme,  $q \in \mathbf{Z}$ ,  $q > 0$ , such that  $x \mapsto x^q$  defines an endomorphism of  $\mathbf{G}_{a,S}$ ,  $(G, T, \alpha)$  and  $(G', T', \alpha')$  two elementary  $S$ -systems,  $f : G \rightarrow G'$  a morphism of  $S$ -groups. The following conditions on  $f$  are equivalent:*

- (i) *The restriction of  $f$  to  $T$  factors as a morphism  $f_T : T \rightarrow T'$  making the diagram commute:*

$$\begin{array}{ccccc} \mathbf{G}_{m,S} & \xrightarrow{\alpha^*} & T & \xrightarrow{\alpha} & \mathbf{G}_{m,S} \\ q \downarrow & & \downarrow f_T & & \downarrow q \\ \mathbf{G}_{m,S} & \xrightarrow{\alpha'^*} & T' & \xrightarrow{\alpha'} & \mathbf{G}_{m,S} \end{array}$$

- (ii) *There exists a (unique) isomorphism of  $\mathcal{O}_S$ -modules*

$$h : (\mathfrak{g}^\alpha)^{\otimes q} \rightarrow \mathfrak{g}'^{\alpha'}$$

*such that*

$$f(\exp(X)) = \exp(h(X^q)), \quad f(\exp(Y)) = \exp(h^\vee(Y^q))$$

*for all  $X \in W(\mathfrak{g}^\alpha)(S')$ ,  $Y \in W(\mathfrak{g}^{-\alpha})(S')$ ,  $S' \rightarrow S$  (it follows that  $f$  also satisfies the equivalent conditions of 3.10).*

One has (ii)  $\Rightarrow$  (i). Indeed, by 3.10, condition (ii) implies  $f \circ \alpha^* = q\alpha'^*$ , hence, by 3.3,  $f|_T$  factors through  $T'$ . It remains to prove that  $\alpha'(f(t)) = \alpha(t)^q$ , which follows at once from the fact that  $f$  induces a morphism of groups  $T \cdot U_\alpha \rightarrow T' \cdot U_{\alpha'}$ .

Let us prove (i)  $\Rightarrow$  (ii). Let  $X \in \Gamma(S, \mathfrak{g}^\alpha)$ ,  $Y \in \Gamma(S, \mathfrak{g}^{-\alpha})$ . Put  $p_+(x) = f(\exp(xX))$  and  $p_-(x) = f(\exp(yY))$ ; these are morphisms of groups

$$p_+, p_- : \mathbf{G}_{a,S} \rightarrow G.$$

Now one has

$$\begin{aligned} \text{int}(\alpha'^*(z))^q(p_+(x)) &= \text{int}(f_T(\alpha^*(z)))(f(\exp(xX))) \\ &= f(\text{int}(\alpha^*(z))(\exp(xX))) \\ &= f(\exp(z^2xX)) = p_+(z^2x). \end{aligned}$$

Applying lemma 1.2 (with  $Q = \alpha'^*(\mathbf{G}_{m,S})$ ), one deduces that there exists a section  $X' \in \Gamma(S, \mathfrak{g}^{\alpha'})$  such that

$$f(\exp(xX)) = p_+(x) = \exp(x^q X').$$

Similarly, there exists a section  $Y' \in \Gamma(S, \mathfrak{g}^{-\alpha'})$  such that

$$f(\exp(yY)) = \exp(y^q Y').$$

Writing now that  $f$  is a morphism of groups, hence respects formula (F), one obtains at once

$$X^q Y^q = (XY)^q = X' Y'.$$

One easily concludes that  $X^q \mapsto X'$  and  $Y^q \mapsto Y'$  define isomorphisms  $h$  and  $h^\vee$  as announced.

**Proposition 3.12.** *Let  $(G, T, \alpha)$  be an elementary  $S$ -system,  $w \in N^\times(S)$ ; put*

$$\Omega_0 = \Omega \cap \text{int}(w^{-1})(\Omega).$$

*Let  $d$  be the function on  $\Omega$  defined by*

$$d(\exp(Y) \cdot t \cdot \exp(X)) = \alpha(t)^{-1} + XY.$$

*Then  $\Omega_0 = \Omega_d$ , and for  $\exp(Y) \cdot t \cdot \exp(X) \in \Omega_0(S')$  one has the following formula (put  $z = d(\exp(Y) \cdot t \cdot \exp(X))$ ):*

$$(\star) \quad \begin{aligned} \text{int}(w)(\exp(Y) \cdot t \cdot \exp(X)) &= \exp(z^{-1} a_\alpha(w)^{-1} X) \\ &\quad \cdot t \alpha^*(z) \cdot \exp(z^{-1} a_\alpha(w) Y). \end{aligned}$$

*Moreover,  $d \circ \text{int}(w) = d^{-1}$ .*

Indeed, one has at once<sup>(30)</sup>

$$\begin{aligned} \text{int}(w)(\exp(Y) \cdot t \cdot \exp(X)) &= \exp(a_\alpha(w) Y) \cdot t \alpha^*(\alpha(t)^{-1}) \cdot \exp(a_\alpha(w)^{-1} X) \\ &= \exp(a_\alpha(w) Y) \cdot \exp(\alpha(t) a_\alpha(w)^{-1} X) \cdot t \alpha^*(\alpha(t)^{-1}). \end{aligned}$$

By 2.1, this is a section of  $\Omega$  if and only if  $1 + \alpha(t)XY$  is invertible, which indeed proves  $\Omega_0 = \Omega_d$ ; then applying formula (F) of loc. cit., one deduces by an immediate calculation the announced formula  $(\star)$ . Finally, it follows from  $(\star)$  that one has

$$\begin{aligned} (d \circ \text{int}(w))(\exp(Y) \cdot t \cdot \exp(X)) &= \alpha(t \alpha^*(z))^{-1} + z^{-2} XY \\ &= z^{-2} (\alpha(t)^{-1} + XY) = z^{-1}, \end{aligned}$$

whence the last assertion.

N. B. One will note that the function  $d$  is independent of the choice of  $w$ .

#### 4. The isomorphism theorem

**Theorem 4.1.** *Let  $S$  be a scheme,  $q \in \mathbf{Z}$ ,  $q > 0$ , such that  $x \mapsto x^q$  is an endomorphism of  $\mathbf{G}_{a,S}$ ,  $(G, T, \alpha)$  and  $(G', T', \alpha')$  two elementary  $S$ -systems. Let*

$$h : (\mathfrak{g}^\alpha)^{\otimes q} \longrightarrow \mathfrak{g}'^{\alpha'} \quad \text{and} \quad h^\vee : (\mathfrak{g}^{-\alpha})^{\otimes q} \longrightarrow \mathfrak{g}'^{-\alpha'}$$

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<sup>(30)</sup>**N.D.E.** The original has been corrected by interchanging  $a_\alpha(w)$  and  $a_\alpha(w)^{-1}$ , and the proof of  $d \circ \text{int}(w) = d^{-1}$  has been expanded.

be two mutually contragredient isomorphisms. Let  $f_T : T \rightarrow T'$  be a morphism of  $S$ -groups making the diagram commute:

$$\begin{array}{ccc} \mathbf{G}_{m,S} & \xrightarrow{q} & \mathbf{G}_{m,S} \\ \alpha^* \downarrow & & \downarrow \alpha'^* \\ T & \xrightarrow{f_T} & T' \\ \alpha \downarrow & & \downarrow \alpha' \\ \mathbf{G}_{m,S} & \xrightarrow{q} & \mathbf{G}_{m,S}. \end{array}$$

There exists a unique morphism of  $S$ -groups  $f : G \rightarrow G'$  extending  $f_T$  and satisfying

$$f(\exp(X)) = \exp(h(X^q))$$

for every  $X \in W(\mathfrak{g}^\alpha)(S')$ ,  $S' \rightarrow S$ . Moreover, this morphism also satisfies

$$f(\exp(Y)) = \exp(h^\vee(Y^q)) \quad \text{and} \quad f(w_\alpha(Z)) = w_\alpha(h(Z^q)),$$

for every  $S' \rightarrow S$  and all  $Y \in \Gamma(S', \mathfrak{g}^{-\alpha})$ ,  $Z \in \Gamma(S', \mathfrak{g}^\alpha)^\times$ .

If  $f : G \rightarrow G'$  extends  $f_T$ , then  $f \circ \alpha^* = (\alpha'^*)^q$ . If in addition  $f$  satisfies the second condition, it also satisfies the other two by 3.10. It follows that  $f$  is determined on  $\Omega$  by the relation

$$f(\exp(Y)t\exp(X)) = \exp(h^\vee(Y^q))f_T(t)\exp(h(X^q)).$$

Since  $\Omega$  is schematically dense in  $G$ , this already proves the uniqueness of  $f$ . To prove its existence, it suffices, by Exp. XVIII 2.3, to prove that the preceding formula defines a “generically multiplicative” morphism from  $\Omega$  to  $G'$ . Now, by 2.4, this amounts to checking that  $\alpha' \circ f = \alpha^q$ , which follows from the fact that  $f$  extends  $f_T$ .

**Scholium 4.2.** One can also interpret 4.1 as follows: consider the category  $\mathcal{E}$  of elementary  $S$ -systems and the category  $\mathcal{D}$  of pairs

$$\left( \mathbf{G}_{m,S} \xrightarrow{\alpha^*} T \xrightarrow{\alpha} \mathbf{G}_{m,S}, \mathcal{L} \right),$$

where  $T$  is a torus,  $\alpha$  and  $\alpha^*$  morphisms of groups such that  $\alpha \circ \alpha^* = 2$ , and  $\mathcal{L}$  an invertible  $\mathcal{O}_S$ -module (the reader will specify the morphisms of the two categories considered). One defines a functor  $\mathcal{E} \rightarrow \mathcal{D}$  by

$$(G, T, \alpha) \mapsto \left( \mathbf{G}_{m,S} \xrightarrow{\alpha^*} T \xrightarrow{\alpha} \mathbf{G}_{m,S}, \mathfrak{g}^\alpha \right).$$

The preceding theorem says that this functor is fully faithful. It is in fact an equivalence of categories, as will be seen in the next section. One already has:

**Corollary 4.3.** If  $q = 1$  and  $f_T$  is an isomorphism, then  $f$  is an isomorphism.

**Corollary 4.4.** If  $q = 1$  and  $f_T$  is faithfully flat with kernel  $Q$  (cf. Exp. IX 2.7), then  $f$  is faithfully flat (quasi-compact) with kernel  $Q$ , hence identifies  $G'$  with  $G/Q$ .

Indeed, if  $f_T$  is faithfully flat with kernel  $Q$ , then

$$Q = \text{Ker}(f_T) \subset \text{Ker}(f_T \circ \alpha') = \text{Ker}(\alpha).$$

Introducing the elementary  $S$ -system  $(G/Q, T/Q, r/Q)$  of 2.13, one is reduced by 2.14 to proving that  $f/Q$  induces an isomorphism from  $G/Q$  onto  $G'$ , which follows at once from 4.3.

## 5. Examples of elementary systems, applications

**5.1.** — Let  $S$  be a scheme,  $\mathcal{L}$  an invertible  $\mathcal{O}_S$ -module. Consider the group  $G_{\mathcal{L}}$  over  $S$  defined by

$$G_{\mathcal{L}}(S') = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \left| \begin{array}{l} a, d \in \mathbf{G}_a(S'), \quad b \in W(\mathcal{L})(S'), \\ c \in W(\mathcal{L}^{-1})(S'), \quad ad - bc \in \mathbf{G}_m(S') \end{array} \right. \right\},$$

equipped with the usual matrix multiplication law. It is locally isomorphic to  $\mathrm{GL}_{2,S}$ . It is therefore a group  $S$ -scheme, affine and smooth over  $S$ , with connected fibers.

**Remark.** Let  $\mathcal{L}'$  and  $\mathcal{L}''$  be two invertible sheaves on  $S$  such that  $\mathcal{L} = \mathcal{L}' \otimes \mathcal{L}''^{-1}$ .<sup>(31)</sup> Then one has an isomorphism of  $S$ -groups:

$$G_{\mathcal{L}} \xrightarrow{\sim} \mathrm{GL}(\mathcal{L}' \oplus \mathcal{L}'')$$

defined as follows: if  $x$  (resp.  $y$ ) is a section of  $\mathcal{L}'$  (resp.  $\mathcal{L}''$ ) over an open subset  $V$  of  $S$ , one has

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} ax + by \\ cx + dy \end{pmatrix}.$$

**5.2.** — One denotes by  $S_{\mathcal{L}}$  the closed subgroup of  $G_{\mathcal{L}}$  defined by  $ad - bc = 1$ . It is also a group  $S$ -scheme, affine and smooth over  $S$ , with connected fibers (isomorphic to  $\mathrm{SL}(\mathcal{L}' \oplus \mathcal{L}'')$  by the preceding isomorphism).

Similarly, consider the morphism  $\mathbf{G}_{m,S} \rightarrow G_{\mathcal{L}}$  defined by  $z \mapsto \begin{pmatrix} z & 0 \\ 0 & z \end{pmatrix}$ . It is a central monomorphism; passing to the quotient, one deduces a group  $P_{\mathcal{L}}$ , smooth and affine over  $S$ , with connected fibers (cf. Exp. VIII 5.7).

One can see that, passing to the quotient from the isomorphism of the preceding remark,  $P_{\mathcal{L}}$  is identified with the group of automorphisms of the projective bundle  $\mathbf{P}(\mathcal{L}' \oplus \mathcal{L}'')$  (cf. EGA, II 4.2.7). One denotes by  $i$  and  $p$  the canonical morphisms

$$S_{\mathcal{L}} \xrightarrow{i} G_{\mathcal{L}} \xrightarrow{p} P_{\mathcal{L}};$$

$i$  is a closed immersion,  $p$  is faithfully flat and affine.

**5.3.** — Consider the morphisms of groups

$$\begin{aligned} t_G : \mathbf{G}_{m,S}^2 &\rightarrow G_{\mathcal{L}}, & t_G(z, z') &= \begin{pmatrix} z & 0 \\ 0 & z' \end{pmatrix}; \\ t_S : \mathbf{G}_{m,S} &\rightarrow S_{\mathcal{L}}, & t_S(z) &= \begin{pmatrix} z & 0 \\ 0 & z^{-1} \end{pmatrix}; \\ t_P : \mathbf{G}_{m,S} &\rightarrow P_{\mathcal{L}}, & t_P(z) &= p(t_G(z, 1)). \end{aligned}$$

These are monomorphisms of groups defining in each group a (split) torus of relative codimension 2. For every  $s \in S$ , let

$$X \in \Gamma(\bar{s}, \mathcal{L} \otimes \bar{s})^{\times};$$

then the section  $\begin{pmatrix} 0 & X \\ X^{-1} & 0 \end{pmatrix}$  of  $G_{\mathcal{L},\bar{s}}$  normalizes  $t_G(\mathbf{G}_{m,\bar{s}}^2)$  and does not centralize it; one then concludes from Exp. XIX 1.6 that  $G_{\mathcal{L}}$  is *reductive*, of *semisimple rank 1*, with *maximal torus*  $t_G(\mathbf{G}_{m,S}^2)$ .

One reasons similarly for  $S_{\mathcal{L}}$  and  $P_{\mathcal{L}}$ , and sees that  $S_{\mathcal{L}}$  (resp.  $P_{\mathcal{L}}$ ) is *reductive*, of *semisimple rank 1*, with maximal torus  $t_S(\mathbf{G}_{m,S})$  (resp.  $t_P(\mathbf{G}_{m,S})$ ).

**5.4.** — Reasoning as usual, one immediately determines the Lie algebra of these various groups and the adjoint action of the chosen maximal torus. Let us do so for  $G_{\mathcal{L}}$ ; this is immediate by Exp. II 4.8:  $\mathrm{Lie}(G_{\mathcal{L}}/S)$  is the Lie algebra of the matrices below:

$$\mathrm{Lie}(G_{\mathcal{L}}/S) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \left| \begin{array}{l} a \text{ and } d \text{ sections of } \mathcal{O}_S, \quad b \text{ a section of } \mathcal{L}, \\ c \text{ a section of } \mathcal{L}^{-1} \end{array} \right. \right\}$$

<sup>(31)</sup>**N.D.E.**  $\mathcal{L}'' \otimes \mathcal{L}'^{-1}$  has been corrected to  $\mathcal{L}' \otimes \mathcal{L}''^{-1}$ , and the following sentence has been expanded.

with the usual bracket; one has

$$\mathrm{Ad}(t_G(z, z')) \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a & zz'^{-1}b \\ z'z^{-1}c & d \end{pmatrix}.$$

Put  $\mathcal{L}ie(G_{\mathcal{L}}/S) = \mathfrak{g}$ . Let  $\alpha_G : t_G(\mathbf{G}_{\mathbf{m},S}^2) \rightarrow \mathbf{G}_{\mathbf{m},S}$  be the character defined by

$$\alpha_G(t_G(z, z')) = zz'^{-1}.$$

One sees at once from the preceding relation that  $\alpha_G$  is a root of  $G_{\mathcal{L}}$  with respect to  $t_G(\mathbf{G}_{\mathbf{m},S}^2)$ , and that the morphism

$$u : \mathcal{L} \rightarrow \mathfrak{g} \quad (\text{resp. } u_- : \mathcal{L}^{-1} \rightarrow \mathfrak{g})$$

defined by  $u(X) = \begin{pmatrix} 0 & X \\ 0 & 0 \end{pmatrix}$  (resp.  $u_-(X) = \begin{pmatrix} 0 & 0 \\ X & 0 \end{pmatrix}$ ) is an isomorphism from  $\mathcal{L}$  onto  $\mathfrak{g}^{\alpha_G}$  (resp. from  $\mathcal{L}^{-1}$  onto  $\mathfrak{g}^{-\alpha_G}$ ).

One has therefore proved that  $(G, t_G(\mathbf{G}_{\mathbf{m},S}^2), \alpha_G)$  is an elementary  $S$ -system.

Putting similarly

$$\alpha_S(t_S(z)) = z^2, \quad \alpha_P(t_P(z)) = z,$$

one proves that  $(S_{\mathcal{L}}, t_S(\mathbf{G}_{\mathbf{m},S}), \alpha_S)$  and  $(P_{\mathcal{L}}, t_P(\mathbf{G}_{\mathbf{m},S}), \alpha_P)$  are elementary systems, and defines isomorphisms of  $\mathcal{L}$  (resp.  $\mathcal{L}^{-1}$ ) with the corresponding direct summands of the Lie algebras of  $S_{\mathcal{L}}$  and  $P_{\mathcal{L}}$ .

**5.5.** — Put  $\exp \begin{pmatrix} 0 & X \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & X \\ 0 & 1 \end{pmatrix}$ . One has thus defined a morphism

$$W(\mathfrak{g}^{\alpha_G}) \rightarrow G_{\mathcal{L}}$$

which induces on Lie algebras the canonical morphism, hence is the unique morphism of this type (1.5). Similarly, put  $\exp \begin{pmatrix} 0 & 0 \\ Y & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ Y & 1 \end{pmatrix}$ . Calculating formula (F) explicitly, one finds

$$\left\langle \begin{pmatrix} 0 & X \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ Y & 0 \end{pmatrix} \right\rangle = XY, \quad \alpha_G^*(z) = \begin{pmatrix} z & 0 \\ 0 & 1/z \end{pmatrix} = t_G(z, z^{-1}).$$

<sup>(32)</sup>The open subset  $N^\times = N_G^\times$  (defined before 3.1) is:

$$N_G^\times(S') = \left\{ \begin{pmatrix} 0 & P \\ Q & 0 \end{pmatrix} \mid P \in W(\mathfrak{g}^\alpha)^\times(S'), Q \in W(\mathfrak{g}^{-\alpha})^\times(S') \right\},$$

the morphism  $w_{\alpha_G}$  (cf. 3.1 (iv)) is given, for every  $X \in W(\mathfrak{g}^\alpha)^\times(S')$ , by

$$w_{\alpha_G}(X) = \begin{pmatrix} 0 & X \\ -X^{-1} & 0 \end{pmatrix};$$

the morphism  $a_{\alpha_G}$  (cf. 3.5) is given by:

$$\text{if } w = \begin{pmatrix} 0 & P \\ Q & 0 \end{pmatrix} \in N_G^\times(S'), \quad \text{then } a_{\alpha_G}(w) = PQ^{-1} \in W((\mathfrak{g}^\alpha)^{\otimes 2})^\times(S'),$$

i.e. for every  $Y \in W(\mathfrak{g}^{-\alpha})^\times(S')$ , one has  $a_{\alpha_G}(w)(Y) = PQ^{-1}Y \in W(\mathfrak{g}^\alpha)^\times(S')$ .

**5.6.** — We leave it to the reader to perform the same calculations in  $S_{\mathcal{L}}$  and  $P_{\mathcal{L}}$ . One finds the same duality formula and the coroots

$$\alpha_S^*(z) = t_S(z), \quad \alpha_P^*(z) = t_P(z^2).$$

Let  $p_T$  be the morphism induced by  $p : G_{\mathcal{L}} \rightarrow P_{\mathcal{L}}$  on  $t_S(\mathbf{G}_{\mathbf{m},S})$ , i.e.

$$p_T(t_S(z)) = t_P(z^2).$$

<sup>(32)</sup>**N.D.E.** The following has been expanded.

One therefore has the commutative diagram:<sup>(33)</sup>

$$\begin{array}{ccccc}
 & & \mathbf{G}_{m,S} & & \\
 & \swarrow \text{id} & \searrow \alpha_S^* & \swarrow \alpha_P^* & \searrow 2 \\
 \mathbf{G}_{m,S} & \xrightarrow{t_S} & t_S(\mathbf{G}_{m,S}) & \xrightarrow{p_T} & t_P(\mathbf{G}_{m,S}) \xrightarrow{t_P} \mathbf{G}_{m,S} \\
 & \searrow 2 & \swarrow \alpha_S & \swarrow \alpha_P & \searrow \text{id} \\
 & & \mathbf{G}_{m,S} & & 
 \end{array}$$

One recognizes in the central part the commutative diagram of 4.1<sup>(34)</sup> relative to the canonical morphism  $p \circ i : S_{\mathcal{L}} \rightarrow P_{\mathcal{L}}$ , which induces a morphism of the preceding elementary  $S$ -systems.

**5.7.** — Let now  $(G, T, \alpha)$  be any elementary  $S$ -system. Consider the commutative diagram:

$$\begin{array}{ccccc}
 & & \mathbf{G}_{m,S} & & \\
 & \swarrow \text{id} & \downarrow \alpha^* & \searrow 2 & \\
 \mathbf{G}_{m,S} & \xrightarrow{\alpha^*} & T & \xrightarrow{\alpha} & \mathbf{G}_{m,S} \\
 & \searrow 2 & \downarrow \alpha & \swarrow \text{id} & \\
 & & \mathbf{G}_{m,S} & & 
 \end{array}$$

Combining the preceding two diagrams, one obtains a commutative diagram:

$$\begin{array}{ccccc}
 & & \mathbf{G}_{m,S} & & \\
 & \swarrow \alpha_S^* & \downarrow \alpha^* & \swarrow \alpha_P^* & \\
 t_S(\mathbf{G}_{m,S}) & \xrightarrow{\alpha^* \circ t_S^{-1}} & T & \xrightarrow{t_P \circ \alpha} & t_P(\mathbf{G}_{m,S}) \\
 & \searrow \alpha_S & \downarrow \alpha & \swarrow \alpha_P & \\
 & & \mathbf{G}_{m,S} & & 
 \end{array}$$

Using 4.1, one therefore has:

**Proposition 5.8.** *Let  $S$  be a scheme,  $(G, T, \alpha)$  an elementary  $S$ -system. Put  $\mathcal{L} = \mathfrak{g}^\alpha$  (and hence  $\mathcal{L}^{-1} = \mathfrak{g}^{-\alpha}$ ).*

(i) *There exists a unique morphism of groups  $f : S_{\mathcal{L}} \rightarrow G$  satisfying the following equivalent conditions:*

$$\begin{aligned}
 (a) \quad & f \begin{pmatrix} z & 0 \\ 0 & z^{-1} \end{pmatrix} = \alpha^*(z), & f \begin{pmatrix} 1 & X \\ 0 & 1 \end{pmatrix} = \exp(X); \\
 (b) \quad & f \begin{pmatrix} 1 & X \\ 0 & 1 \end{pmatrix} = \exp(X), & f \begin{pmatrix} 1 & 0 \\ Y & 1 \end{pmatrix} = \exp(Y); \\
 (c) \quad & f \begin{pmatrix} 1 & X \\ 0 & 1 \end{pmatrix} = \exp(X), & f \begin{pmatrix} 0 & X \\ -X^{-1} & 0 \end{pmatrix} = w_\alpha(X).
 \end{aligned}$$

(ii) *There exists a unique morphism of groups  $g : G \rightarrow P_{\mathcal{L}}$  satisfying*

$$g(t) = \begin{pmatrix} \alpha(t) & 0 \\ 0 & 1 \end{pmatrix}, \quad g(\exp(X)) = p \begin{pmatrix} 1 & X \\ 0 & 1 \end{pmatrix}.$$

<sup>(33)</sup>**N.D.E.** Where  $t_S$  and  $t_P$  are isomorphisms.

<sup>(34)</sup>**N.D.E.** With  $q = 1$ .

Moreover, one has

$$g(\exp(Y)) = p \begin{pmatrix} 1 & 0 \\ Y & 1 \end{pmatrix}, \quad g(w_\alpha(X)) = p \begin{pmatrix} 0 & X \\ -X^{-1} & 0 \end{pmatrix}.$$

The morphism  $g$  is faithfully flat quasi-compact with kernel  $\text{Ker}(\alpha) = \underline{\text{Centr}}(G)$ , and  $g \circ f$  is the canonical morphism  $S_{\mathcal{L}} \rightarrow P_{\mathcal{L}}$ .

Note that conditions (b) of (i) give an explicit description of the duality between  $\mathfrak{g}^\alpha$  and  $\mathfrak{g}^{-\alpha}$ .

**Corollary 5.9.** *Let  $(G, T, \alpha)$  be an elementary  $S$ -system. The subgroups  $T \cdot U_\alpha$ ,  $T \cdot U_{-\alpha}$ ,  $U_\alpha$  and  $U_{-\alpha}$  are closed.*

Since  $U_\alpha$  is a closed subgroup scheme of  $T \cdot U_\alpha$ , it suffices to check this for the latter. By Noether's theorem (Exp. IV 5.3.1 and 6.4.1), it suffices to prove that  $(T \cdot U_\alpha)/\text{Ker}(\alpha)$  is a closed subgroup of  $G/\text{Ker}(\alpha)$ . By 5.8, one is therefore reduced to proving that the subgroup of  $P_{\mathcal{L}}$  (or of  $G_{\mathcal{L}}$ , which amounts to the same thing by another application of Noether's theorem) defined by  $c = 0$  is closed, which is trivial.

Consequently, the morphisms  $\exp$  of theorem 1.5 (i) are closed immersions.

N. B. The corollary also follows from the fact that  $T \cdot U_\alpha$  and  $T \cdot U_{-\alpha}$  are “Borel subgroups” of  $G$  (cf. Exp. XII 7.10).

**5.10.** — Let  $\mathcal{L}$  be an invertible  $\mathcal{O}_S$ -module and

$$\mathbf{G}_{m,S} \xrightarrow{\alpha^*} T \xrightarrow{\alpha} \mathbf{G}_{m,S}$$

a diagram of groups<sup>(35)</sup> such that  $\alpha \circ \alpha^* = 2$ . Let  $R$  be the maximal torus of  $\text{Ker}(\alpha)$  and  $K = \alpha^{*-1}(R)$ . Then  $K$  is a subgroup of multiplicative type of  $\mathbf{G}_{m,S}$ ; by  $\alpha \circ \alpha^* = 2$ , it is even a subgroup of  $\mu_{2,S}$ . In particular, the morphism

$$K \rightarrow S_{\mathcal{L}}, \quad z \mapsto \begin{pmatrix} z & 0 \\ 0 & z^{-1} \end{pmatrix}$$

is central. One therefore has a central monomorphism of groups:

$$K \rightarrow R \times S_{\mathcal{L}}, \quad z \mapsto \left( \alpha^*(z), \begin{pmatrix} z & 0 \\ 0 & z^{-1} \end{pmatrix} \right).$$

Consider the group  $G = (R \times S_{\mathcal{L}})/K$  obtained by passing to the quotient. It is a group affine and smooth over  $S$ , with connected fibers. It is immediate that the sequence

$$1 \rightarrow K \rightarrow R \times t_S(\mathbf{G}_{m,S}) \rightarrow T \rightarrow 1$$

where  $u(x, t_S(z)) = x\alpha^*(z)$  is exact. The image of  $R \times t_S(\mathbf{G}_{m,S})$  in  $G$  is therefore a torus  $T'$  isomorphic to  $T$ . One now shows without difficulty that if  $\alpha'$  is the character of  $T'$  deduced from  $\alpha$  by the preceding isomorphism,  $(G, T', \alpha')$  is an elementary  $S$ -system, that  $\mathfrak{g}^{\alpha'}$  is isomorphic to  $\mathcal{L}$ , and that  $\alpha'^*$  is obtained from  $\alpha^*$  by the isomorphism  $T \xrightarrow{\sim} T'$ . One has therefore constructed an elementary  $S$ -system  $(G, T', \alpha')$  such that the corresponding object  $(\mathbf{G}_{m,S} \xrightarrow{\alpha'^*} T' \xrightarrow{\alpha'} \mathbf{G}_{m,S}, \mathfrak{g}^{\alpha'})$  of the category  $\mathcal{D}$  defined in 4.2 is isomorphic to  $(\mathbf{G}_{m,S} \xrightarrow{\alpha^*} T \xrightarrow{\alpha} \mathbf{G}_{m,S}, \mathcal{L})$ . One has therefore proved:

**Theorem 5.11.** *With the notations of 4.2, the functor*

$$(G, T, \alpha) \mapsto (\mathbf{G}_{m,S} \xrightarrow{\alpha^*} T \xrightarrow{\alpha} \mathbf{G}_{m,S}, \mathfrak{g}^\alpha)$$

*is an equivalence of categories between  $\mathcal{E}$  and  $\mathcal{D}$ .*

## 6. Generators and relations for an elementary system

<sup>(35)</sup>N.D.E.  $T$  being a torus.

**6.1.** — Let  $S$  be a scheme,  $(G, T, \alpha)$  an elementary  $S$ -system. Let  $X \in W(\mathfrak{g}^\alpha)^\times(S)$  and  $u = \exp(X)$ ; we saw in 3.8 that the element  $w = w_\alpha(X)$  satisfies, in particular, the relation

$$(wu)^3 = e.$$

<sup>(36)</sup>Let  $s_\alpha$  denote the automorphism of  $T$  induced by  $\text{int}(w)$ ; by theorem 3.1 (iii), for every  $S' \rightarrow S$  and  $t \in T(S')$ , one has

$$s_\alpha(t) = \text{int}(w)(t) = t \cdot \alpha^*(\alpha(t)^{-1}).$$

**Theorem 6.2.** *Let  $H$  be a sheaf of  $S$ -groups for (fppf). Let*

$$f_T : T \rightarrow H, \quad f_\alpha : U_\alpha \rightarrow H$$

*be morphisms of groups and  $h \in H(S)$  a section of  $H$ . For there to exist a morphism of groups (necessarily unique)*

$$f : G \rightarrow H$$

*extending  $f_T$  and  $f_\alpha$  and satisfying  $f(w) = h$ , it is necessary and sufficient that the following conditions hold:*

- (i) *For every  $S' \rightarrow S$ , every  $t \in T(S')$  and every  $x \in U_\alpha(S')$ , one has*

$$(1) \quad f_T(t)f_\alpha(x)f_T(t)^{-1} = f_\alpha(txt^{-1}) = f_\alpha(x^{\alpha(t)}).$$

*(in other words,  $f_T$  and  $f_\alpha$  extend to a morphism of groups from the semidirect product  $T \cdot U_\alpha$  to  $H$ ).*
- (ii) *For every  $S' \rightarrow S$  and every  $t \in T(S')$ , one has*

$$(2) \quad hf_T(t)h^{-1} = f_T(s_\alpha(t)) = f_T(t \cdot \alpha^*(\alpha(t)^{-1})).$$
- (iii) *One has the two relations in  $H(S)$ :*

$$(3) \quad h^2 = f_T(\alpha^*(-1)),$$

$$(4) \quad (hf_\alpha(u))^3 = e.$$

*Proof.* Let us write  $U_\alpha$  and  $U_{-\alpha}$  additively, and their vector structure multiplicatively. If  $f$  satisfies the conditions of the statement, one necessarily has, for every  $y \in U_{-\alpha}(S')$ ,

$$f(y) = f(w^{-1}yww^{-1}) = hf_\alpha(w^{-1}yw)h^{-1}.$$

Let therefore  $f_{-\alpha} : U_{-\alpha} \rightarrow H$  be the morphism defined by

$$(*_1) \quad f_{-\alpha}(y) = hf_\alpha(w^{-1}yw)h^{-1}.$$

It is a morphism of groups. On the other hand,  $f$  is determined on the big cell  $\Omega$  by

$$f(ytx) = f_{-\alpha}(y)f_T(t)f_\alpha(x).$$

This shows the uniqueness of  $f$ ; since the conditions of the statement are manifestly necessary, let us show that they are sufficient.

By (4), one has

$$hf_\alpha(u)h^{-1}h^2 = f_\alpha(-u)h^{-1}f_\alpha(-u).$$

Now, by (3) and (1),  $h^2 = h^{-2}$  commutes with  $f_\alpha(-u)$ , which gives

$$hf_\alpha(u)h^{-1} = f_\alpha(-u)hf_\alpha(-u).$$

But, by definition,  $hf_\alpha(u)h^{-1} = f_{-\alpha}(wuw^{-1})$ ; by 3.7, since  $u = \exp(X)$  and  $w = w_\alpha(X)$ , one has

$$(*_2) \quad wuw^{-1} = -\tilde{u},$$

where  $\tilde{u}$  denotes the element paired with  $u$ . One therefore obtains:

$$(*_3) \quad f_{-\alpha}(-\tilde{u}) = f_\alpha(-u)hf_\alpha(-u).$$

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<sup>(36)</sup>**N.D.E.** The following sentence has been added.

Let now  $t$  be a section of  $T$  over a variable  $S' \rightarrow S$ . Let  $\text{int}(f_T(t))$  act on the preceding formula. On the left-hand side, one obtains<sup>(37)</sup>

$$\begin{aligned} f_T(t)f_{-\alpha}(-\tilde{u})f_T(t)^{-1} &= f_T(t)hf_{\alpha}(u)h^{-1}f_T(t)^{-1} \\ &= h(h^{-1}f_T(t)h)f_{\alpha}(u)(h^{-1}f_T(t)^{-1}h)h^{-1} \\ &= hf_T(s_{\alpha}(t))f_{\alpha}(u)f_T(s_{\alpha}(t))^{-1}h^{-1} \\ &= hf_{\alpha}(\alpha(s_{\alpha}(t))u)h^{-1} \end{aligned}$$

by (2) and (1); then, since  $s_{\alpha}(t) = t \cdot \alpha^*(\alpha(t)^{-1})$  and  $\alpha \circ \alpha^* = 2$ , this equals

$$hf_{\alpha}(\alpha(t)^{-1}u)h^{-1}.$$

Finally, by  $(*_1)$  and  $(*_2)$ , one has

$$hf_{\alpha}(\alpha(t)^{-1}u)h^{-1} = f_{-\alpha}(\alpha(t)^{-1}uwu^{-1}) = f_{-\alpha}(-\alpha(t)^{-1}\tilde{u}).$$

The right-hand side of  $(*_3)$  gives

$$f_{\alpha}(-\alpha(t)u) \cdot f_T(t)hf_T(t)^{-1}h^{-1} \cdot h \cdot f_{\alpha}(-\alpha(t)u),$$

and since  $hf_T(t)^{-1}h^{-1} = f_T(s_{\alpha}(t^{-1})) = f_T(t \cdot \alpha^*(\alpha(t)))$ , this equals

$$f_{\alpha}(-\alpha(t)u) \cdot f_T(\alpha^*(\alpha(t))) \cdot h \cdot f_{\alpha}(-\alpha(t)u).$$

Comparing the two expressions obtained, one obtains

$$f_{-\alpha}(-\alpha(t)^{-1}\tilde{u}) = f_{\alpha}(-\alpha(t)u) \cdot f_T(\alpha^*(\alpha(t))) \cdot h \cdot f_{\alpha}(-\alpha(t)u).$$

Since  $\alpha : T \rightarrow \mathbf{G}_{m,S}$  is faithfully flat and  $H$  is a separated presheaf, one may replace  $-\alpha(t)^{-1}$  by an arbitrary section of  $\mathbf{G}_{m,S}$  and obtains:

**Lemma 6.2.1.** *For every  $z \in \mathbf{G}_m(S')$ ,  $S' \rightarrow S$ , one has*

$$f_{-\alpha}(z\tilde{u}) = f_{\alpha}(z^{-1}u) \cdot f_T(\alpha^*(-z^{-1})) \cdot h \cdot f_{\alpha}(z^{-1}u).$$

Let now  $x, y \in \mathbf{G}_a(S')$ ,  $S' \rightarrow S$ ; suppose  $y$  and  $(1 + xy)$  invertible. First applying the lemma with  $z = y$ , one obtains<sup>(38)</sup>

$$f_{\alpha}(xu)f_{-\alpha}(y\tilde{u}) = f_{\alpha}((x + y^{-1})u) \cdot f_T(\alpha(-y^{-1})) \cdot h \cdot f_{\alpha}(y^{-1}u).$$

Now  $x + y^{-1} = y^{-1}(1 + xy)$ . Applying the lemma with  $z = y/(1 + xy)$ , one finds

$$\begin{aligned} f_{\alpha}((x + y^{-1})u) &= f_{-\alpha}\left(\frac{y}{1 + xy}\tilde{u}\right) f_{\alpha}(-(x + y^{-1})u) \\ &\quad \cdot h^{-1} \cdot f_T\left(\alpha^*\left(\frac{-y}{1 + xy}\right)\right). \end{aligned}$$

Substituting this in the preceding equality, one obtains

$$\begin{aligned} f_{\alpha}(xu)f_{-\alpha}(y\tilde{u}) &= f_{-\alpha}\left(\frac{y}{1 + xy}\tilde{u}\right) f_{\alpha}(-(x + y^{-1})u) \\ &\quad \cdot h^{-1} \cdot f_T(\alpha^*(1 + xy)^{-1}) \cdot h \cdot f_{\alpha}(y^{-1}u). \end{aligned}$$

Since  $h^{-1}f_T(t)h = f_T(s_{\alpha}(t))$  by (2) (note that  $s_{\alpha}^2 = \text{id}$ ), and since  $s_{\alpha} \circ \alpha^* = -\alpha^*$  (cf. 6.2.1), this equals

$$f_{-\alpha}\left(\frac{y}{1 + xy}\tilde{u}\right) f_{\alpha}(-(x + y^{-1})u) \cdot f_T(\alpha^*(1 + xy)) \cdot f_{\alpha}(y^{-1}u).$$

Finally, since for all  $x' \in U_{\alpha}(S')$  and  $z \in \mathbf{G}_m(S')$ , one has

$$f_{\alpha}(x')f_T(\alpha^*(z)) = f_T(\alpha^*(z))f_{\alpha}(z^{-2}x'),$$

<sup>(37)</sup>**N.D.E.** The following has been corrected.

<sup>(38)</sup>**N.D.E.** The following calculations have been expanded.

one obtains

$$\begin{aligned} f_\alpha(xu)f_{-\alpha}(y\tilde{u}) &= f_{-\alpha}\left(\frac{y}{1+xy}\tilde{u}\right) \cdot f_T(\alpha^*(1+xy)) \cdot f_\alpha\left(\left(\frac{-y^{-1}(1+xy)}{(1+xy)^2} + y^{-1}\right)u\right) \\ &= f_{-\alpha}\left(\frac{y}{1+xy}\tilde{u}\right) \cdot f_T(\alpha^*(1+xy)) \cdot f_\alpha\left(\frac{x}{1+xy}u\right). \end{aligned}$$

One has therefore proved:

**Lemma 6.2.2.** *Let  $S' \rightarrow S$ . If  $a \in U_\alpha(S')$ ,  $b \in U_{-\alpha}^\times(S')$ , and  $1+ab \in \mathbf{G}_m(S')$ , one has*

$$f_\alpha(a)f_{-\alpha}(b) = f_{-\alpha}\left(\frac{b}{1+ab}\right) f_T(\alpha^*(1+ab)) f_\alpha\left(\frac{a}{1+ab}\right).$$

By schematic density, this formula remains valid when  $b \in U_{-\alpha}(S')$ ,  $1+ab$  still being invertible. Consider then the morphism

$$f : \Omega \rightarrow H$$

defined by  $f(ytx) = f_{-\alpha}(y)f_T(t)f_\alpha(x)$ .

It follows at once from 6.2.2, condition 6.2 (i), and formula  $(F')$  of 2.4, that if  $g, g' \in \Omega(S')$  and  $gg' \in \Omega(S')$ , one has  $f(gg') = f(g)f(g')$ . By Exp. XVIII 2.3 (iii) and 2.4,<sup>(39)</sup> there therefore exists a morphism of groups  $G \rightarrow H$  extending  $f$ . Denote it also by  $f$ ; it answers the question; indeed, it suffices to prove that  $f(w_\alpha) = h$ . Now  $w_\alpha = u \cdot (-\tilde{u}) \cdot u$ , whence, by  $(*)_3$ :<sup>(40)</sup>

$$f(w_\alpha) = f_\alpha(u)f_{-\alpha}(-\tilde{u})f_\alpha(u) = h.$$

**Remark 6.3.** We shall supplement these results in Exp. XXIII 3.5.

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<sup>(39)</sup>**N.D.E.** Note that each geometric fiber of  $G$  is connected, for example by 1.1.

<sup>(40)</sup>**N.D.E.** The original has been simplified by invoking  $(*)_3$ .